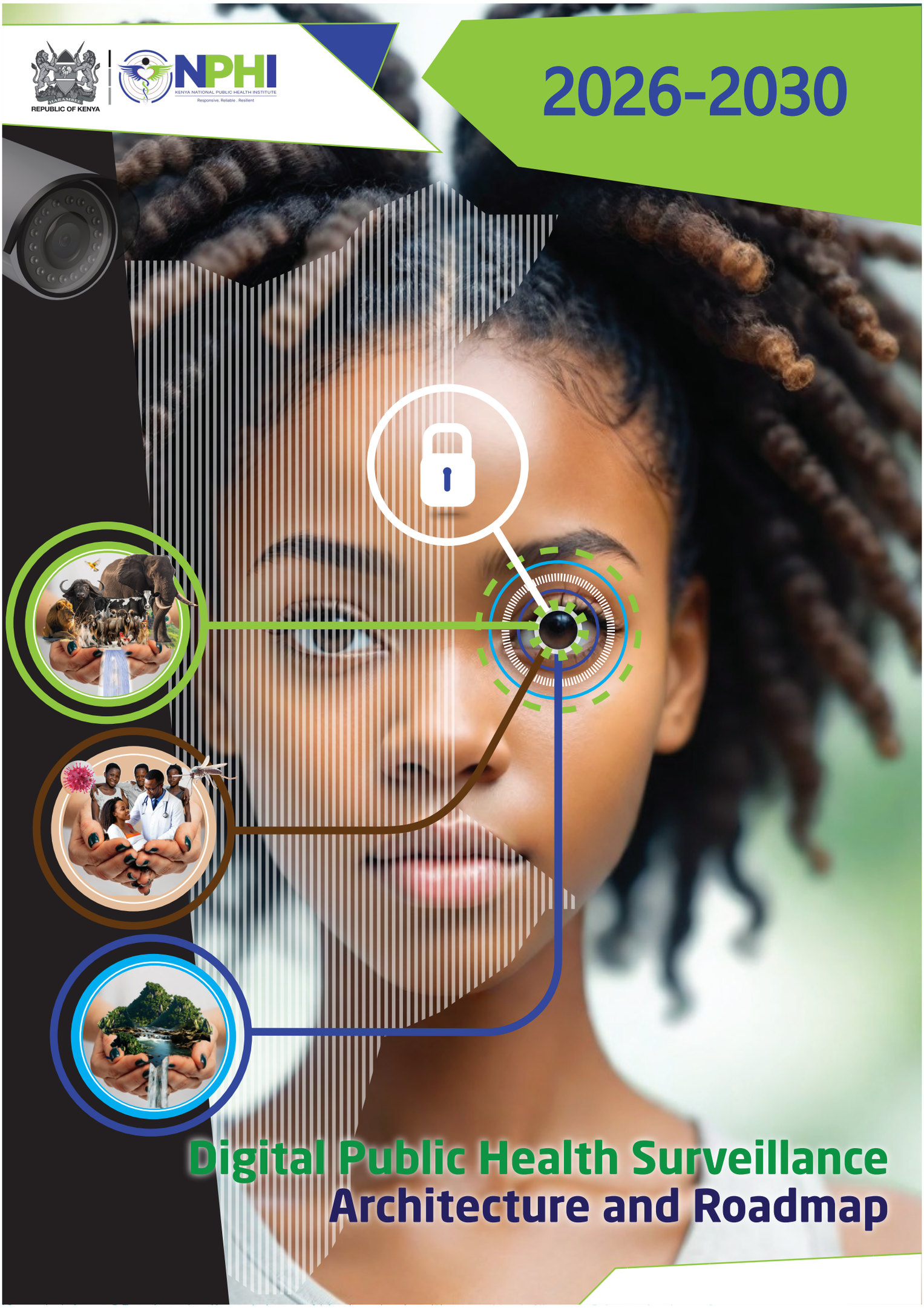
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2026-2030



Digital Public Health Surveillance Architecture and Roadmap

Digital Public Health Surveillance
Architecture and Roadmap

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Foreword



Public Health Surveillance in Kenya is led by the Ministry of Health (MOH) through the Kenya National Public Health Institute (KNPHI). The KNPHI, in collaboration with various divisions within the Ministry, operates under the guidance of the International Health Regulations (IHR) and the Integrated Disease Surveillance and Response (IDSR) guidelines, along with other policy frameworks. The Ministry's divisions and directorates work closely together, as well as with other ministries such as the Ministry of Agriculture and Livestock Development, Ministry of Water, Sanitation and Irrigation and the Ministry of Environment, Climate Change & Forestry, to coordinate public health surveillance and response efforts. The KNPHI is responsible for developing guidelines and standard operating procedure (SOP) to govern public health surveillance in Kenya.

The COVID-19 pandemic underscored the need to enhance collaboration and coordination among the entities responsible for public health surveillance and response, further strengthening the One Health approach. It also revealed the limitations of Kenya's traditional, paper-based public health surveillance systems. In response, KNPHI has crafted a roadmap to transition to a digitally enabled public health surveillance system, aligning with the broader digital transformation of Kenya's health sector.

The establishment of the Kenya National Public Health Institute (KNPHI), through Legal Notice No. 14 of January 21, 2022, marked a significant step in this transition. The KNPHI is mandated with strengthening and promoting public health systems by consolidating the management and coordination of activities related to the prevention, early detection and management of public health emergencies and other events of concern.

This document outlines the current challenges and strategies for transforming Kenya's public health surveillance system into a digitized future. The KNPHI, in collaboration with MoH directorates (the Directorate of Public Health, the Digital Health Agency, and the Directorate of Health Financing, Digital Health, Policy, and Research) and the Digital Health Agency (DHA), will play a pivotal role in guiding this transition. Digital health technologies will be key in enabling efficient data collection, driving automated analysis and interpretation of public health data, and facilitating timely and effective responses to emerging health threats.

Kenya's journey toward digital public health surveillance aims to create a more resilient, responsive, and integrated health system, leveraging technology to safeguard public health in an increasingly interconnected world.

Dr. Kamene Kimenye 'ndc'(K).
Ag. Director General
Kenya National Public Health Institute (KNPHI)

Preface

The COVID-19 pandemic demonstrated, on a global scale, the critical importance of robust and responsive public health surveillance systems. In Kenya, this crisis revealed vulnerabilities in our traditional methods of monitoring and responding to public health threats, exposing the limitations of manual, paper-based surveillance systems in managing modern health emergencies. As Kenya continues to grow and its public health challenges evolve, it has become evident that embracing digital innovation is not just an option but a necessity.

The Digital Public Health Surveillance Architecture and Roadmap marks a key milestone in Kenya's health sector journey towards modernization. It outlines a strategic shift from conventional public health surveillance models to a digitized future, where technology becomes an essential driver of health intelligence and response. This transition is aligned with the broader digital transformation underway across the health sector, aimed at creating an integrated, agile, and data-driven public health system capable of addressing both present and future challenges.

The strategy reflects the collaborative efforts of various public health institutions, ministries, and stakeholders, all united in the vision of a health system that harnesses the power of technology to protect the well-being of the nation. At its core, this document is a roadmap that will guide our approach to digitizing public health surveillance and improving our capacity to prevent, detect, and respond to health threats in real-time.

As we embark on this journey, we acknowledge the importance of strong governance, clear policies, and effective coordination across all levels of the health system. The Ministry of Health through Kenya National Public Health Institute (KNPHI), and other key entities, will lead the implementation of this strategy, ensuring that it delivers on its promise of transforming Kenya's public health landscape. It is our hope that this strategy will not only build a more resilient health system but also inspire confidence in Kenya's ability to meet the demands of a changing world.

Acknowledgements

The development of the **Digital Public Health Surveillance Architecture and Roadmap** would not have been possible without the collaborative efforts and dedication of numerous individuals, institutions, and partners. We would like to extend our deepest appreciation to all those who contributed their expertise, time, and commitment to this important initiative.

First and foremost, we acknowledge the leadership of the KNPHI for spearheading the development of this strategy. Special thanks go to the Directorate of Public Health Research and Informatics for its pivotal role in coordinating the design and implementation of the digitalization roadmap, and for its continued efforts in strengthening public health systems across the country.

We also extend our gratitude to the Ministry of Health especially the Directorate of Public Health, and the Directorate of Health Informatics and Research alongside the Digital Health Agency, for their technical guidance and insight throughout this process. Their expertise in digital health and informatics has been invaluable in shaping a forward-looking and innovative strategy.

We are especially grateful to the various governmental ministries, including the ministries responsible for Livestock, Wildlife, Environment, Water and Sanitation., as well as county health departments, for their collaboration and support in ensuring that the strategy aligns with the broader One Health approach. Their input has helped to create a comprehensive strategy that addresses the full spectrum of public health surveillance needs in Kenya.

Our heartfelt thanks go to the development partners with a special call out to US CDC and Palladium who have helped shape and develop the document, international health organizations, and stakeholders from the private sector and civil society, whose support and technical assistance were instrumental in enriching the strategy with global best practices and innovations. Their partnership is critical in helping Kenya realize its vision of a digitized and resilient public health system.

Finally, we would like to acknowledge the tireless efforts of all public health professionals working on the frontlines. Their commitment to safeguarding the health of Kenyans, particularly during the challenges posed by the COVID-19 pandemic, has been the driving force behind the creation of this strategy. It is their work that inspires us to continue striving for a more efficient, responsive, and technologically advanced public health system.

We are confident that through this collective effort, Kenya will move closer to realizing a future where digital solutions enhance public health surveillance and ensure the well-being of all its citizens.

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Abbreviations

AFENET	Africa Field Epidemiology Network
AIDS	Acquired Immunodeficiency Syndrome
APHL	Association of Public Health Laboratories
API	Application Programming Interface
ARIS	Animal Resource Information System
AU-IBAR	African Union Inter-African Bureau of Animal Resources
CHAI	Clinton Health Access Initiative
CDC	US Centers for Disease Control and prevention
CIF	Case Investigation Form
CHT	Community Health Toolkit
CLTS	Community Led Total Sanitation
CoVID-19	Coronavirus disease 2019
DVO	District Veterinary Officer
EARS	Emergency Alerts and Response System
EBS	Events Based Surveillance
ECHIS	Electronic Community Health Information System
EOC	Emergency Operations Center
ETL	Extract Transform Load
FETP	Field Epidemiology Training Program
FHIR	Fast Healthcare Interoperability Resources
HIE	Health Information Exchange
HIV	Human Immunodeficiency Virus
IBS	Indicator Based Surveillance
ICAP	International Center for AIDS Care and Treatment Programs
IDSR	Integrated Disease Surveillance and Response
IHR	International Health regulations
ILI	Influenza-Like Illness

- KABS Tool — Kenya Animal Bio Surveillance Tool
- KNPHI — Kenya National Public Health Institute
- KPI — Key Performance Indices
- LMIS — Lab Management Information System
- MDR — Multi-Drug Resistance
- MOH — Ministry of Health
- MPT — Mobile Phone Technology
- NDSA — National Digital Stewardship Alliance
- NNT — Neonatal Tetanus
- NTD — Neglected Tropical Diseases
- OIE — Office International des Epizooties
- ODF — Open Defecation Free
- OpenHIM — Open Health Information Mediator
- PWDs — Persons with Disabilities
- RRT — Rapid Response Team
- RUSH RT-MIS — Rural Urban Sanitation Hygiene Real Time Monitoring Information System
- SARI — Severe Acute Respiratory Infection
- SARS — Severe Acute Respiratory Syndrome
- SARS-COV2 — Severe Acute Respiratory Syndrome Coronavirus-2
- SDG — Sustainable Development Goal
- STI — Sexually Transmitted Infections
- TB — Tuberculosis
- UNICEF — United Nations Children's Fund
- UPI — Unique Person Identifier
- VBDs — Vector Borne Diseases
- WAHIS — World Animal Health Information System
- WHO — World Health Organization
- XDR — Extremely Drug Resistant

Executive Summary

Recent challenges in and threats to global public health, particularly with regards to infectious diseases, have resulted in the need for solid health security infrastructures worldwide. As a signatory to the World Health Organisation's (WHO's) International Health Regulations (IHR) and Global Health Security Agenda (GHSA), Kenya is committed to the establishment of robust preparedness and response mechanisms, conducted in compliance with suggested guidelines, to guarantee population health.

The goal of this Public Health Digitization Architecture and Roadmap is to outline a strategy for both government and society to collaborate to ensure that Kenya's health security measures are implemented in a way that efficiently handles and guards against global health threats, bolstering the nation's ongoing surveillance efforts, investing in its health workforce, and ensuring excellent laboratory services.

Currently, disease surveillance in Kenya faces several challenges, such as inadequate and siloed systems, lack of standardisation and privacy challenges, all which limit outbreak response effectiveness.

This document seeks to address these challenges by presenting: a strategic plan for the monitoring of and response to public health issues through the collection and reporting of priority disease data; initiatives for the surveillance of other diseases; steps that can be taken to integrate disease surveillance transitioning manually-collected data to electronic platforms and strengthening these platforms to eliminate the need for paper-based surveillance; and the steps required to implement and scale up the process. It incorporates IDSR as well as other surveillance initiatives, aiming to broaden and strengthen Kenya's Public Health Digitization Architecture. Notably, it also outlines Kenya's Zoonotic Disease Unit's efforts to study the interactions between animal and human health under the One Health approach, to further enhance health surveillance.

The Public Health Digitization Architecture and Roadmap is presented to various state and non-state stakeholders in Kenya's health sector.

1

Public Health Surveillance and Response in Kenya

1.1 Introduction

1.1.1 Background

The International Health Regulations (IHR) (2005) [1] and the Global Health Security Agenda (GHSA) are fundamental initiatives that aim to enhance Kenya's defences against global infectious disease threats. The IHR, established by the World Health Organisation (WHO), serves as a binding legal framework that delineates member states' responsibilities in preventing, detecting, and responding to international public health emergencies. As a signatory to the IHR, Kenya is committed to constructing a robust health security infrastructure aligned with these regulations. Similarly, Kenya actively engages in the GHSA—a collaborative, multi-sectoral approach aimed at enhancing global health security. The GHSA framework fosters partnerships among governments, international organisations, private sectors, and civil society to address gaps in health security capacities. Within Kenya, the GHSA acts as a catalyst, reinforcing preparedness and response mechanisms through initiatives spanning surveillance systems, laboratory networks, workforce training, and emergency capabilities.

A whole-of-government and whole-of-society approach is crucial to the IHR and GHSA collaboration in Kenya. This entails integrating health security measures across diverse sectors, including agriculture, transportation, and education, recognising the interconnected nature of global health challenges. While the IHR provides the regulatory foundation, the GHSA serves as an operational framework, ensuring that Kenya not only complies with international standards, but also actively endeavours to build a resilient health system capable of effectively managing health threats.

Within the context of IHR, Kenya has established National IHR Focal Points and designated competent authorities responsible for the implementation and coordination of health security measures. These focal points facilitate communication and collaboration between Kenya and the global health community, ensuring the timely sharing of information on potential public health events. Ongoing attempts to track and assess the development of core capacities serve as another indication of Kenya's dedication to IHR. This has been made possible through the Joint External Evaluation (JEE), which was carried out in 2017 in cooperation with WHO, has made this possible. For each of the 13 IHR core capacities, a rating was supplied by the JEE report [2]. Kenya is continually working to strengthen its surveillance capacities through various initiatives.

GHSA complements these IHR efforts by providing a platform for collaboration and resource mobilisation. In Kenya, GHSA initiatives have led to the development of strategic plans, the establishment of emergency operations centers, and the enhancement of cross-border collaboration to address regional health security challenges. The combination of IHR and GHSA in Kenya reflects a proactive and integrated approach to global health security, emphasising the importance of collective action in the face of emerging infectious diseases.

Public health surveillance in Kenya has undergone significant evolution over the years, driven by the country's commitment to public health and its response to emerging threats. As a geographical hub, Kenya faces a diverse range of health challenges, including infectious diseases and non-communicable diseases. Effective public health surveillance is crucial for timely detection, response, and mitigation of health threats. This document provides a comprehensive overview of the development, current state, challenges, and prospects of public health surveillance in Kenya.

Historical Development

The roots of public health surveillance in Kenya can be traced back to the colonial era, when rudimentary health systems were established primarily for controlling infectious diseases such as malaria and tuberculosis. With the attainment of independence in 1963, Kenya gradually strengthened its public health infrastructure, including surveillance capabilities. This infrastructure initially focused on the eradication of communicable diseases through targeted interventions.

In the 1990s, the HIV/AIDS pandemic presented a new and formidable challenge. Kenya responded by intensifying surveillance efforts, leading to the establishment of the Integrated Disease Surveillance and Response (IDSR) system. IDSR aimed to integrate surveillance for various diseases and improve reporting, analysis, and response mechanisms. Over the years, Kenya has continued to refine and expand its disease surveillance systems in response to emerging infectious and non-communicable threats.

1.1.2. IDSR in Kenya

The Integrated Disease Surveillance and Response (IDSR) system in Kenya is a comprehensive public health strategy designed to strengthen the country's ability to detect, monitor, and respond to various diseases in a timely and effective manner. The WHO introduced this integrated approach to disease surveillance to enhance global health security through the WHO Afro Initiative of 1998, which WHO Regional Office for Africa (AFRO) to guide health development efforts in the African region.

In Kenya, the IDSR system was officially adopted in 2000 from a specific strategy within the WHO Afro Initiative that focused on improving the capacity of countries in the African region to detect, report, and respond to public health threats. By adopting of IDSR, Kenya made a shift from traditional vertical disease-specific surveillance programs to a more holistic and integrated approach. The primary goal of IDSR is to improve the country's capacity to detect and respond to disease outbreaks promptly, thereby reducing morbidity, mortality, and the socio-economic impact of infectious diseases.

Key components of the IDSR system in Kenya include:

Case Detection and Reporting: IDSR emphasises the early detection of cases through a network of health facilities, laboratories, and community health workers. Health workers are trained to recognise and report specific diseases promptly.

Laboratory Surveillance: IDSR promotes the integration of laboratory data into the surveillance system, ensuring reporting accuracy and timely diagnosis of diseases. This is crucial for confirming outbreaks and guiding appropriate public health responses.

Data Management and Analysis: IDSR stresses the importance of effective collection, management, and analysis of surveillance data. This involves the use of technology and information systems to streamline reporting processes and enhance the ability to detect trends and anomalies.

Communication and Coordination: IDSR encourages the establishment of robust communication channels and coordination mechanisms among various stakeholders involved in disease surveillance and response. This includes collaboration between national, regional, and local health authorities, as well as international partners.

Training and Capacity Building: Ongoing training and capacity-building efforts are essential to ensure the success of IDSR. Health workers at various levels are trained to use the system effectively, and continuous professional development programs are implemented to keep the workforce informed about new surveillance techniques and response strategies.

The IDSR system in Kenya has played a vital role in improving the country's ability to respond to public health threats. It has been particularly instrumental in the early detection of outbreaks, such as those caused by emerging, and re-emerging infectious diseases. By integrating multiple disease surveillance efforts into a unified system, Kenya is better positioned to respond effectively to emerging health challenges, contributing to overall improvements in health outcomes and the protection of public health.

1.1.3 Management of IDSR in Kenya

Disease surveillance in Kenya is organised and managed by KNPHI in collaboration with partners at multiple levels of the healthcare system.

1. **Kenya National Public Health Institute (KNPHI):** KNPHI plays a central role in coordinating and overseeing disease surveillance and response activities. It sets guidelines, and standards for surveillance practices and response initiatives across the country in collaboration with the state department for Public Health.

2. **County Health Departments:** County health departments are involved in implementing policies and managing disease surveillance at the local level. They collect data, monitor health trends, and respond to outbreaks within their jurisdictions.

3. **Kenya Field Epidemiology and Laboratory Training Program (KFELTP):** KFELTP is a training program that builds capacity in field epidemiology. It plays a crucial role in training epidemiologists and health professionals to strengthen disease surveillance and response at both national and county levels.

4. **National Public Health Laboratories:** Laboratory services are an integral part of disease surveillance. The National Public Health Laboratories in Kenya contribute to the identification and confirmation of diseases through diagnostic testing.

5. **International Partners:** Kenya collaborates with international organisations such as WHO, the US Centers for Disease Control and Prevention (CDC), FAO, Global Fund and other partners to enhance disease surveillance capabilities, receive technical support, and benefit from global health initiatives.

1.2. Current State of Public Health Surveillance and Response

Kenya has made significant advancements in enhancing disease surveillance, adopting a multifaceted approach. The Ministry of Health oversees surveillance activities, collaborating with various stakeholders, including international organisations, non-governmental organisations, and research institutions. Key components of the current disease surveillance system include:

1. **Integrated Disease Surveillance and Response (IDSR):** The cornerstone of Kenya's surveillance efforts, IDSR integrates surveillance for a wide range of diseases, covering not only infectious diseases but also non-communicable diseases.

2. **Laboratory Surveillance:** Kenya has strengthened its laboratory infrastructure to support surveillance activities. The National Public Health Laboratories (NPHL) and Kenya Medical Research Institute (KEMRI) play a vital role in laboratory diagnostics and research, contributing to a more robust surveillance system.

3. **Health Information Systems:** The country has invested in digital health information systems, facilitating real-time data collection, analysis, and reporting. This has improved the efficiency and accuracy of disease surveillance. Some of the digital health solutions, such as Electronic Community Health Information Systems, have been enhanced to support Events-Based Surveillance in communities.

4. **International Collaboration:** Kenya actively collaborates with international partners, such as the WHO and the CDC, to align its surveillance practices with global standards.

1.2.1 Digital Health and Public Health Surveillance in Kenya

Prior to the COVID-19 outbreak in December 2019, Kenya relied on traditional, predominantly paper-based surveillance approaches. While these traditional surveillance methods have been effective in monitoring public health events and controlling outbreaks, they have several limitations. They are often siloed, slow, and error-prone, and lead to underreporting of cases, hindering the ability of the Ministry of Health to quickly detect and respond to emerging health threats. These limitations can potentially be addressed by transitioning to digital, more real-time surveillance approaches. This transition to a functional, digitally enabled disease surveillance system is essential for improving the speed and accuracy of disease monitoring, resulting in timely detection and response to health problems.

In Kenya, the COVID-19 pandemic presented a unique opportunity for the Ministry of Health to begin adoption of modern technologies for disease surveillance. Driven by the demand for more real-time, accurate data on the evolving COVID-19 pandemic, there emerged an array of digital tools for COVID-19 surveillance and management. Due to the unprecedented nature of the pandemic, this transition to e-surveillance was characterised by uncoordinated adoption and implementation of unstandardised, siloed digital solutions designed for specific surveillance efforts. As such, they lacked the flexibility required by the broader disease surveillance ecosystem.

The efficiency gains and challenges experienced during the COVID-19 period highlighted the need to have concerted efforts towards the adoption of scalable, standards-based technology solutions for near-real-time, electronic disease surveillance and response in the country.

Digital health technologies play a pivotal role in strengthening disease surveillance, improving reporting rates, and enhancing overall health security in Kenya. The value of digital health in disease surveillance has been demonstrated in:

1. **Transition to DHIS2:** In 2016, Kenya shifted from a standalone web-based surveillance system to the more sustainable and integrated District Health Information System 2 (DHIS2). This transition allowed for better data management, real-time reporting, and improved surveillance metrics.[3]

2. **Joint External Evaluation (JEE):** Kenya first conducted a Joint External Evaluation (JEE) of the International Health Regulations (2005) core capacities in 2017, in collaboration with WHO and other partners. The 2017 JEE provided a detailed assessment of Kenya's capacity to prevent, detect, and respond to public health threats, identifying both strengths and critical areas for improvement. Its findings informed the development of the National Action Plan for Health Security (NAPHS) 2019–2023, which guided investments in preparedness, surveillance, and response systems. Building on this progress, Kenya undertook a second JEE mission from 9–13 September 2024, which reaffirmed notable advancements while outlining priority actions to further align with the International Health Regulations (IHR 2005). The 2024 JEE emphasized the need to expand surveillance through nationwide rollout of event-based and community-based surveillance, integration of event- and indicator-based systems, and digitalization of data collection up to the fa-

cility level. Importantly, the 2024 JEE recognized Kenya's progress in developing integrated digital surveillance systems and encouraged further alignment of these systems to the Kenya National Public Health Institute (KNPHI) mandate and the national e-disease surveillance roadmap. This includes ensuring interoperability across human, animal, food safety, and environmental health sectors to strengthen early warning, risk assessment, and response.

3. **National Action Plan for Health Security:** Kenya's NAPHS (2023) was developed based on the findings of the Joint External Evaluation (JEE) conducted in 2017. Its primary purpose is to consolidate all health security capacity-strengthening initiatives into a unified framework, serving as a reference point for stakeholders engaged in health security activities. The NAPHS complements other national disaster and emergency response plans and initiatives to ensure coordinated action. Currently, Kenya is in the process of updating the NAPHS for the 2025–2027 period, guided by insights from the JEE 2024. This updated plan will be pivotal in aligning national health security priorities with the revised International Health Regulations (IHR, 2005) amendments adopted globally in 2024 and formally agreed upon by Kenya in 2025.

4. **Digital Biosurveillance:** Researchers in 2021 explored the potential of digital biosurveillance in detecting zoonotic diseases like Rift Valley fever, anthrax, rabies, brucellosis, and trypanosomiasis. By monitoring large volumes of open-source data, digital surveillance can flag potential health threats promptly. [5]

5. **County-Level Capacities:** Critical reviews assess county-level capacities in operationalizing event-based surveillance and evidence-based health decision-making. These reviews examine gaps, health data in emergency contexts, and comprehensive reporting across community, private health facilities, and faith-based health facilities. [6]

6. **Community Health Digitisation:** Kenya's National Community Health Digitisation Strategy 2020-2025 emphasises the role of digital tools in community health. [7]

In 2022, MOH, in partnership with CDC-Kenya, collaborated through implementing partners to introduce an electronic Integrated Disease Surveillance and Response (e-IDSR). The e-IDSR initiative aims to achieve the following objectives:

1. **National Data Linkage, Repository, & Analytics:** This component focuses on establishing a robust system for data linkage, archiving, and analytics. It supports reporting and facilitates data utilisation for public health preparedness and response.

2. **Public Health Data Automation:** By automating reporting processes, the e-IDSR enables the efficient use of individual-level data for **population health analytics.** This automation streamlines reporting and enhances data quality.

3. **Health Information Exchange (HIE):** The e-surveillance architecture seeks to establish an HIE that improves patient-level services across the health sector. This exchange of health information enhances coordination and decision-making.

4. **Capacity Strengthening:** The initiative focuses on **informatics and digital health** training for the public health workforce. Equipping professionals with these skills ensures effective operation of information systems aligned with core capabilities.

5. **Interoperable Systems:** The e-surveillance architecture aims to coordinate efforts among **global, regional, and national partners.** [8]

This e-surveillance roadmap document outlines a strategic plan for enhancing public health monitoring and response. It complements the Integrated Disease Surveillance and Response (IDSR) by establishing a broad-based strategy for digitising public health surveillance and response workflows that incorporates other surveillance initiatives, such as surveillance at points of entry, environmental surveillance, syndr+omic surveillance, passive disease surveillance, and disease surveillance in domestic and wild animals. The goal is to broaden the surveillance ecosystem and improve the Public Health Architecture through digitisation.

1.2.2. Milestones in the evolution of IDSR in Kenya

The Ministry of Health has achieved several milestones in the implementation of IDSR in Kenya since the WHO-RC resolution on IDSR in 1998. Some of these efforts include development of solutions and repositories. Below is a summary of key milestones towards adoption and implementation of IDSR in Kenya.

IDSR Implementation Milestone

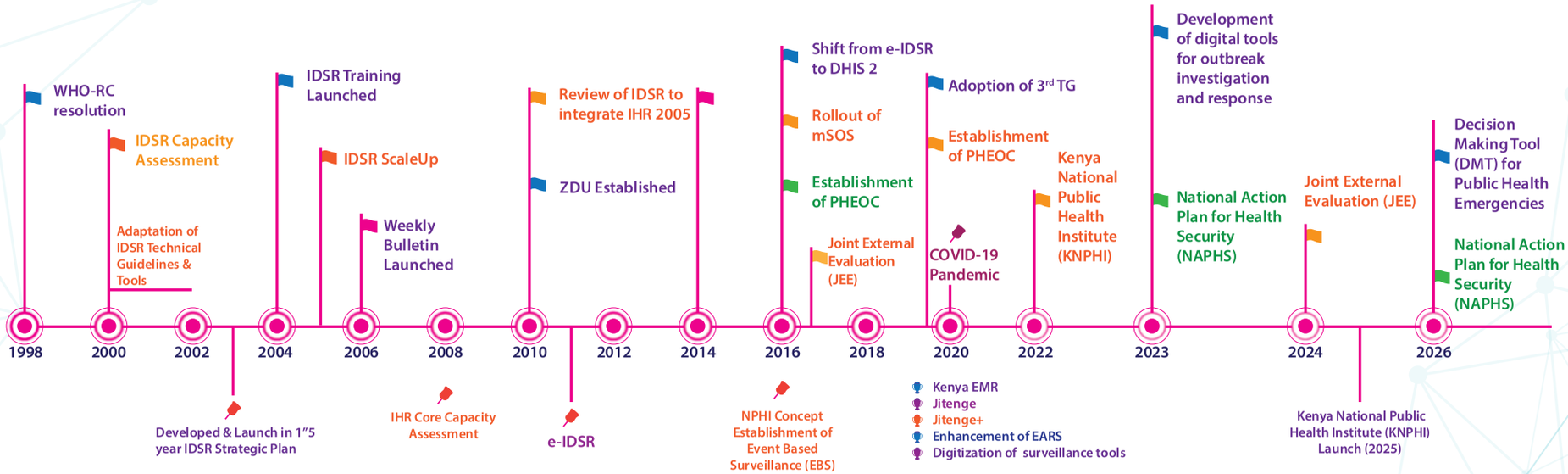


Figure 1: IDSR Implementation Milestones

Pre-Covid19 Period:

- **1998**: IDSR strategy adopted by WHO Afro member states.
- **2000 - 2002**: The first ever IDSR capacity assessment was done and the first IDSR 5-year strategic. plan launched. This led to the adoption of IDSR standards and guidelines.
- **2005**: IDSR was scaled up to all districts for 18 diseases.
- **2006**: The first weekly epidemiological bulletin on the was published.
- **2007**: The EPI-Info database system was hosted at Afya House, supported by the CDC.
- **2008**: There was an attempt to implement phones for health, but this was not successful.
- **2011:** MOH partnered with CHAI and Strathmore University to develop e-IDSR.
- **2013:** There was a shift from e-IDSR to DHIS 2 due to the costly maintenance of the platform
- **2017:** Event Based Surveillance was established.

COVID-19 Period:

- 2020: COVID-19 hit the country and several systems were developed and rolled out to support COVID-19 surveillance and response efforts, including Contact Tracing and Quarantine Management. These solutions included:
- COVID-19 Tracker (CHT and Kenya EMR)
- Jitenge for quarantine management, Jitenge+ for Traveller Surveillance
- Emergency Alerts and Response System (EARS) was customized for COVID-19 Surveillance
- Laboratory Repository and Laboratory Information Management system for remote logging, sample management
- Regional Electronic Cargo & Driver Tracking System
- ChanjoKE: COVID-19 Vaccine Information Management System and Vaccine Certificates
- COVID-19 Dashboard

Most tools developed during the COVID-19 pandemic were non-interoperable, bespoke tools not designed for the surveillance of other public health events.

Post-Covid19:

- 2022: MOH and disease surveillance implementing partners begin work on e-IDSR system
- 2022: Three-year funding through America Rescue Plan Act (ARPA) through CDC for Data Modernization Initiative (DMI) towards
  - Establishment of a Surveillance Repository,
  - Establishment of an Outbreak Responses digital solution,
  - Development of an e-Surveillance Architecture and Roadmap,
  - Development of an interoperable Disease surveillance ecosystem based on the architecture,
  - A Health Data Governance Framework that supports ethical sharing of Surveillance data,
  - Introduction of Informatics training in FELTP curriculum to enable e-disease surveillance.

2

Current State: Situation analysis of Public Health Surveillance and Response



2.1. Integrated Disease Surveillance and Response in Kenya

The Ministry of Health (MOH) in Kenya acknowledges the pivotal role of disease surveillance in the planning, execution, monitoring, and evaluation of public health practice. Collaborating with partners, the MOH has established clear indicators for disease surveillance and developed outcomes to guide the performance of individual actors. The Integrated Disease Surveillance and Response (IDSR) strategy and its implementation in Kenya demonstrate functionality, encompassing the following practices:

- Use of standard IDSR case definitions for the identification and reporting of priority diseases.
- Adoption of surveillance data to provide signals of outbreaks at higher levels and trigger localized actions.
- Investigation and confirmation of suspected outbreaks or public health events,

- Laboratory confirmations when necessary.
- Analysis and interpretation of data from outbreak investigations and routine monitoring of priority diseases.
- Implementation of appropriate public health responses based on data.
- Facilitation of feedback within and across levels of the health system.
- Evaluation and enhancement of the performance of surveillance and response systems.

2.1.1. Priority Diseases Under IDSR

As of April 2024, Kenya's IDSR guidelines had 48 IDSR priority diseases. The IDSR priority diseases were categorised into three groups: epidemic-prone diseases; diseases targeted for eradication or elimination; and other major diseases, events or conditions of public health significance. The criteria for the selection of these conditions included diseases causing high morbidity and mortality, diseases with high epidemic potential, diseases requiring international surveillance, diseases having a national intervention program, and diseases that are identifiable through simple case investigations. Additionally, IDSR in Kenya was dynamic and flexible, enabling the country to monitor emerging and re-emerging diseases and other public health events.

The Ministry of Health relies on different approaches for public health surveillance. Events-Based Surveillance is implemented at community level through applications such as mDharura and the Electronic Community Health Information System (ECHIS). EBS has also been implemented in several health facilities and at points of entry. Indicator Based Surveillance is the most widely used disease surveillance approach at the facility level. Case reporting is a requirement for some IDSR diseases such as polio and measles.

Epidemic-prone diseases, conditions or events which require immediate reporting	Diseases targeted for eradication or elimination	Other major diseases, events or conditions of public health importance
Acute haemorrhagic fever syndrome*	Dracunculiasis (Guinea Worm Disease)	Acute jaundice
Anthrax	Leprosy	Adverse events following immunization (AEFI)
Bacterial Meningitis	Rabies	Diabetes mellitus (new cases)
Chikungunya	Lymphatic filariasis	Diarrhea with dehydration in children of less than 5 years of age
Cholera	Malaria	Epilepsy
Dengue fever	Measles	HIV/AIDS (new cases)
Dysentery (Shigella)	Neonatal tetanus	Hypertension (new cases)
Malaria	Poliomyelitis***	Injuries (road traffic accidents)
Plague	Onchocerciasis	Extreme Malnutrition in children under 5 years of age
SARIs**	Trachoma	Maternal deaths
SARS	Human African Trypanosomiasis	Non-neonatal tetanus
Typhoid fever	*** Disease specified by IHR (2005) for immediate notification	Perinatal deaths
Yellow fever		Severe pneumonia in children of less than 5 years of age
Measles		STIs
NNT		Schistosomiasis
Dracunculiasis		Soil transmitted helminths
Maternal death		Trypanosomiasis
Zika virus		Tuberculosis (new cases)
Also:		MDR/XDR Tuberculosis
A cluster of deaths in the community.		
A cluster of unwell people or animals with similar symptoms	Diseases or events of international concern	
	Human influenza due to a new subtype***	
	SARS***	
	Smallpox***	
	Zika virus disease	
	Yellow fever	
	Any public health event of international or national concern (infectious, zoonotic, food borne,	
*Ebola, Marburg, Rift Valley, Lassa, Crimean Congo, West Nile Fever	*** Disease specified by IHR (2005) for immediate notification	
**National programmes may wish to add Influenza-like illnesses to their priority disease list		

Figure 2: Priority Diseases Under IDSR

2.1.2 Reporting Requirements

Kenya's IDSR reporting guidelines classifies IDSR diseases in three broad categories and stipulates three reporting requirements and frequencies for each category of priority diseases based on their magnitude of impact:

- Immediate reporting (Within 24 hours): for epidemic diseases and those targeted for eradication.
- Weekly IDSR reporting: for other major diseases, events, or conditions of public health importance.
- Monthly IDSR reporting: for additional significant health events.

2.1.3. Data Collection Process and Tools

The current IDSR reporting system comprises of various levels of data capture.

2.1.3.1. Community Level:

Event-based surveillance (EBS) is a core componenEvent-Based Surveillance (EBS) is the systematic detection, collection, triaging, verification, and interpretation of unstructured or ad hoc information—referred to as signals or events—that may indicate acute public health threats. In Kenya, EBS complements Indicator-Based Surveillance (IBS) under the Integrated Disease Surveillance and Response (IDSR) framework.

The Community Health Strategy (2020–2025) integrates EBS into community health structures to strengthen early warning systems, risk assessment, and rapid response capabilities. This integration ensures that communities play a proactive role in identifying and reporting unusual health events, thereby enhancing national health security.

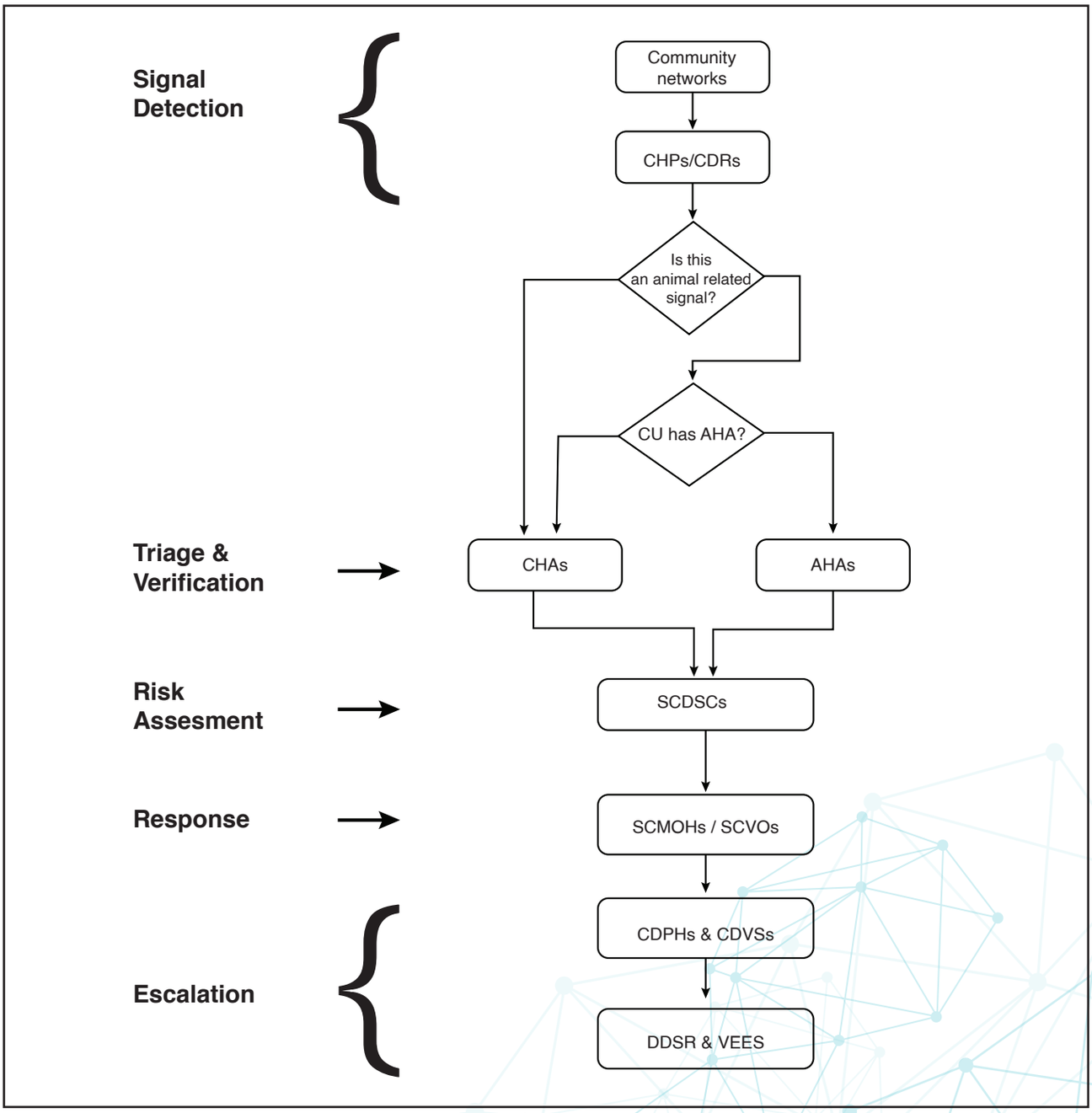


Figure 3a: EBS Data Flow / flow of information

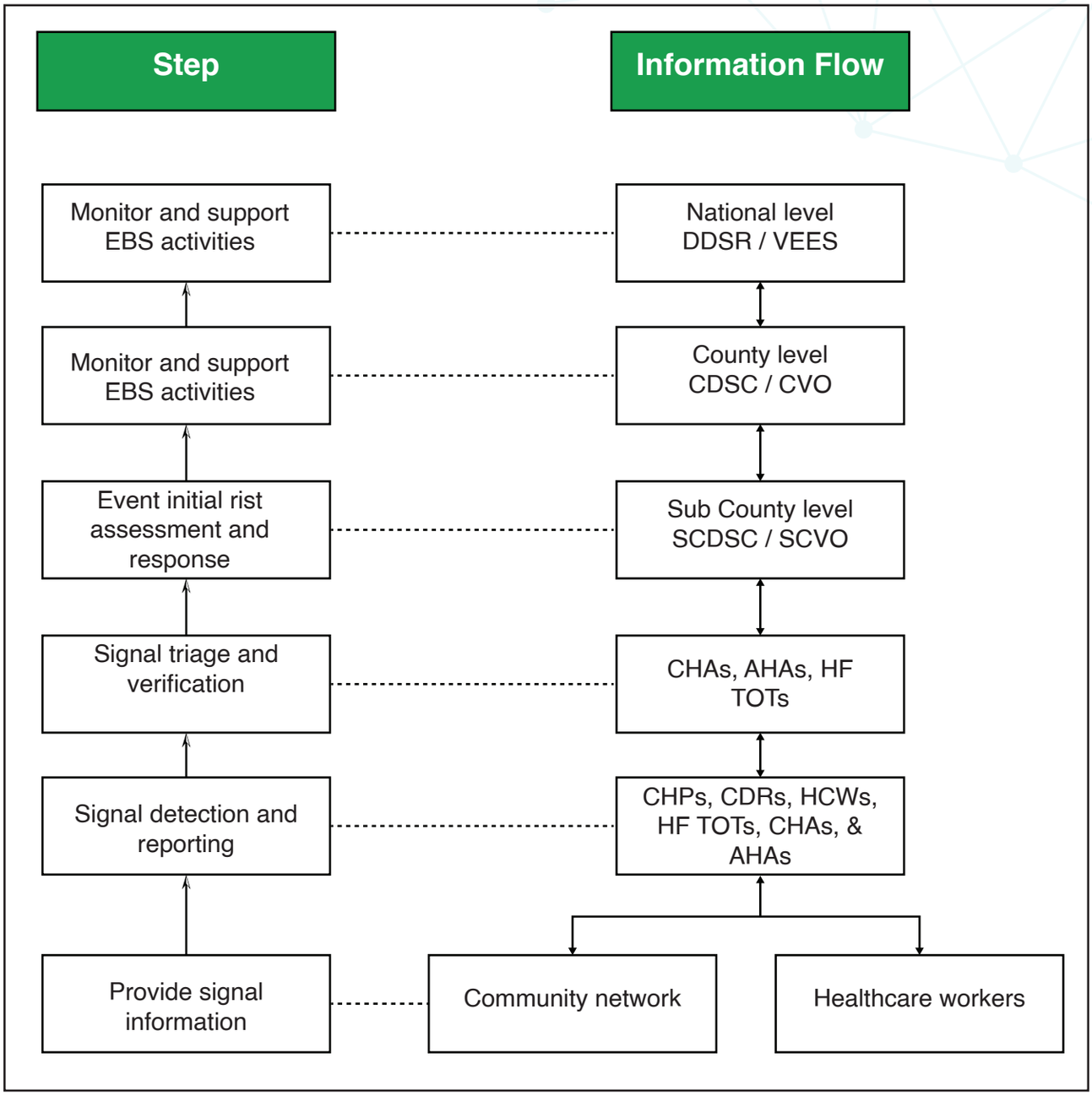


Figure 3b: Community EBS and Health facility EBS Workflow

2.1.3.2. Facility Level:

Integrated disease surveillance occurs at this level. Suspected cases of communicable diseases identified in health facilities or referred from communities are recorded and reported using MOH approved tools. In most cases, IDSR data is captured using paper-based tools, Digital tools (Excel, EMRs) are used in some instances. The collated data is physically transported to sub-county offices where data is entered into the national reporting platform, Kenya Health Information Systems (KHIS).

Indicator-Based Surveillance (IBS) at health facility level

IBS consists of the systematic and routine collection of structured data, mainly from health facility-based, formal sources, and is the conventional form of surveillance in many countries, including Kenya.

The tools used in data collection and reporting at facility level are outlined in the table below.

Tool	MOH Identifier	Purpose
Outpatient Morbidity Summary sheet	(MOH 204 A) (MOH 204 B)	Recording and reporting suspected cases of diseases under surveillance for under-fives and over-fives. These registers are currently paper based in most facilities. Some digital versions can be found in a few facilities.
Laboratory Register	(MOH 240)	Recording Laboratory samples and, tests conducted and the test results in laboratories across the country.
Case Investigation Form	(MOH 502)	The form collects specific demographic and epidemiological details on the suspected cases and is used to accompany the sample to the lab for testing. This form is generic and accommodates several diseases, including cholera, AFP, NNT, and measles.
Acute Flaccid Paralysis Case Validation Form	(MOH 502 B)	Detailed case investigation/ validation form for AFP suspected cases. Collects demographic information, clinical information, and specimen collection information.
Line List of Confirmed Cases	(MOH 503)	Once cases have been confirmed, the MOH 503 is used to prepare a list (Line List) of the confirmed cases (MOH 503) This is mostly manual, and, in some cases is presented as a Microsoft Excel spreadsheet. Each outbreak disease is maintained on a separate line list.
Weekly Aggregate report	(MOH 505)	The report captures weekly occurrences of diseases including outbreaks for conditions that are weekly reportable.

Table 1: IDSR Data Collection and Reporting Tools

Sentinel surveillance of respiratory diseases in selected facilities.

Currently, respiratory disease surveillance for ILI/ SARI is conducted using a sentinel surveillance approach. About ten (10) Health Facilities designated as Sentinel Surveillance sites evenly distributed across the country are used for surveillance of acute respiratory diseases. Disease Surveillance Officers stationed in these sites routinely abstract data for suspected cases of ILI/SARI from clinical care records and record them in a netbook. The data syncs automatically from the netbooks to a national repository where it is analysed and used for respiratory disease surveillance.

2.1.3.3. Sub-County and County Level:

Reporting of indicators to KHIS is conducted by Disease surveillance Officers (DSO) and Health Records and Information Officers (HRIO) stationed at facility or County levels. Most health facilities generate MOH summary reports at site level. Physical copies of these summaries are submitted to the sub-county level. At this level, Disease Surveillance Officers assisted by Health Records and Information Officers (HRIOs) manually receive aggregate summaries from health

facilities and enter the data manually into KHIS. Some large health facilities (level 4,5 and 6) may have their own HRIOs or SOs who have been assigned credentials to KHIS and can enter the data at facility level.

2.1.3.4. National Level:

Disease surveillance reports submitted from facilities are visualised in KHIS at the national level. The Division of Disease Surveillance and Response (DDSR) dashboard in KHIS is a provides a centralised view of weekly disease occurrence, patterns, and trend analysis for IDSR diseases. Data collected from Sentinel Surveillance Health Facilities is also analysed at this level.

IDSR Data Flow

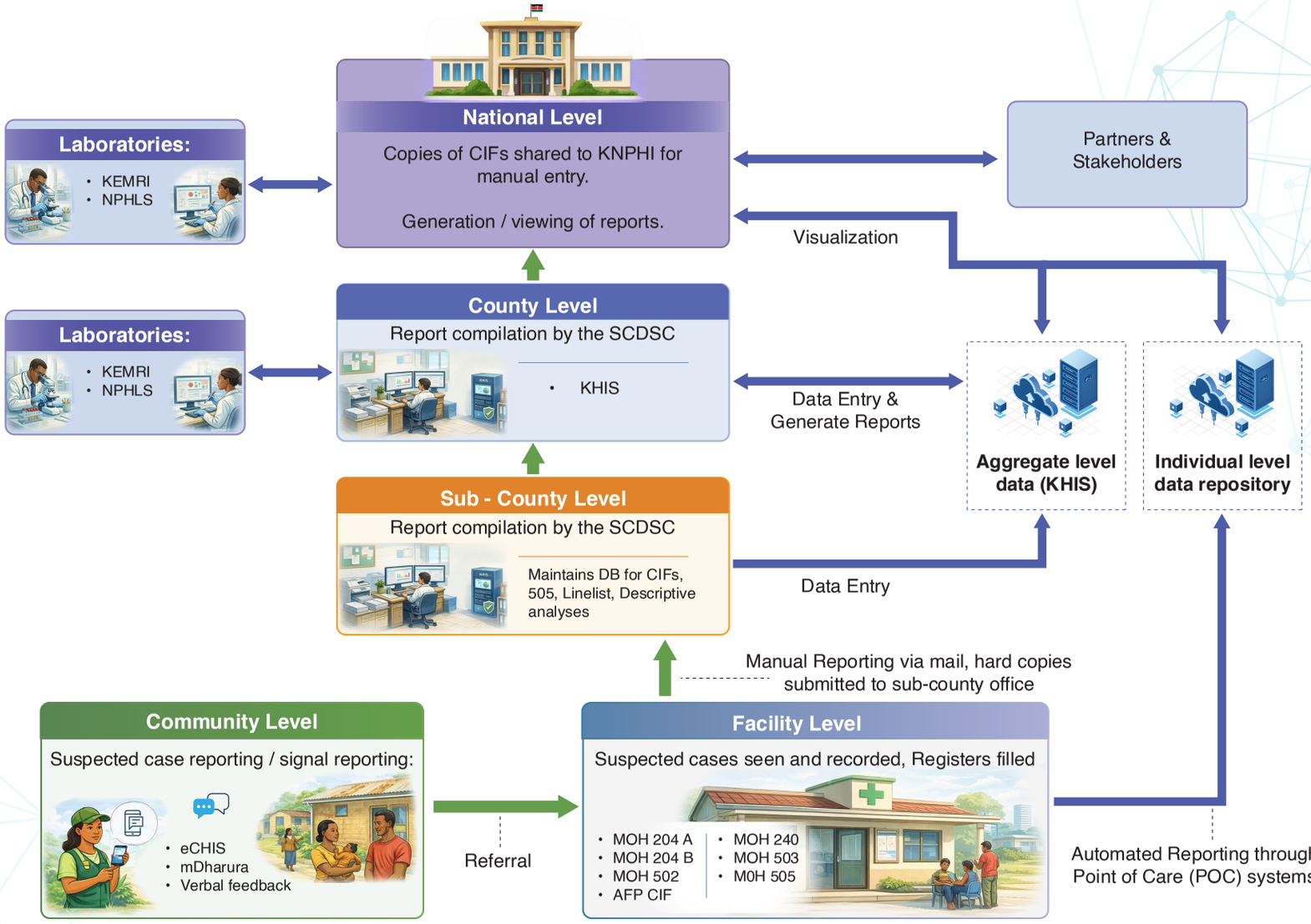


Figure 4: IDSR Data Flow

2.1.4. Limitations of Current IDSR

KNPHI currently implements several disease surveillance initiatives; the KNPHI leverages several digital health tools to meet its surveillance objectives. While it has largely been successful, gaps and challenges remain that need to be addressed. A notable gap is the unintegrated digital surveillance solutions, which would support the routine and outbreak disease monitoring and response that is critical at the national level.

These challenges have resulted in missed opportunities for early case detection, delays in response initiatives and slow resolution of outbreaks. Some of the key challenges encountered are:

i. **Siloed systems that do not exchange data:** Several digital health solutions that support various disease surveillance initiatives are in place. These systems operate in silos and, in some cases, collect similar data leading to duplication of efforts in system development, maintenance. The multiple solutions introduce data collection burden to service providers. Moreover, the multiplicity of data collection points which do not integrate at some point leads to slow or missed opportunities in outbreak response.

ii. **Inadequate systems in place to support extensive outbreak response:** To effectively respond to disease outbreaks, the Ministry of Health requires a robust digital solution that can be easily configured to collect surveillance data for multiple diseases without the intervention of technical resources.

iii. **Limited implementation of digital community surveillance solutions:** The current solution in place that supports community Events-based surveillance is deployed in only six counties, limiting the surveillance initiative scope. The roll out of Electronic Community Health Information System (ECHIS) with EBS forms presents a great opportunity for harmonizing EBS data collection and for strengthening community EBS Data in ECHIS is still not available for disease surveillance. The data residing in ECHIS should be availed to County and Sub-County teams to guide response to public health threats.

iv. **KHIS System downtime and user access:** The KHIS platform, which is used for aggregate reporting by the facilities, experiences frequent downtimes, which hinder timely tracking and generation of reports. Additionally, MOH has limited rights to create users, and lengthy access request processes lead to inefficiency.

v. **System hosting and data ownership does not lie with MOH:** Currently, several systems that support surveillance initiatives are hosted and supported by implementing partners. The Ministry of Health has neither access to nor control over some of the data. This leads to a laborious data request process just for routine surveillance. Additionally, once the partners exit, there are no clear handover mechanisms. This leads to loss of data as some of this data is stored in private repositories.

vi. **Lack of standardisation in person identification:** Currently, the surveillance program has no defined unique identifier. This has made it difficult to uniquely identify cases and link laboratory and clinic data. The roll-out of a National Unique Person Identifier is a priority.

vii. **Variables collected by different systems are non-standard:** With the proliferation of several systems which are developed by different partners, data linkage and analysis has become difficult since due to non- conformance to a common set of standards and guidelines.

viii. **Privacy and data protection challenges.** Lack of knowledge about and imbalance in application of the Data Protection Act often leads to denial of data requests, limited data sharing and non-adherence to privacy and protection provisions.

ix. **Lack of feedback:** Lack of feedback to generators of disease surveillance data and public health teams working in health facilities, counties and in some cases county level leads to under-utilisation of public health surveillance data for evidence-based decision making.

2.2. Other Public Health Surveillance Initiatives in Kenya

Overview

The Division of Disease Surveillance and Response (DDSR) in the Ministry of Health coordinates disease surveillance efforts. Several systems, and tools are designed to collect disease surveillance data that is used to prevent, detect, and ensure rapid response to outbreaks of infectious diseases and other public health threats.

Despite successes in IDSR, several challenges still exist. These challenges include fragmentation of surveillance efforts, duplication, lack of integration, data quality issues, and limited coverage in some cases. To address these gaps, opportunities for automation and digitisation have been identified and recommended.

Below is a summary of the surveillance initiatives under the Directorate of Public Health and recommendations for addressing current challenges with disease surveillance and response systems in use.

2.2.1. Public Health Surveillance at Points of Entry.

Under IHR, member states must comply with the legal requirements set out for designated Points of Entry. [10] Public health surveillance at points of entry is designed to detect and prevent the introduction or spread of infectious diseases and other public health threats through designated points of entry. Public Health Surveillance at points of entry involves ongoing collection of surveillance data on occurrence and prevention of priority diseases among travellers at designated points of entry. Disease surveillance at points of entry is implemented at designated airports, maritime ports, and land border crossings. As of 2024, port health surveillance activities were implemented in 23 out of the 33 gazetted points of entry. These include eight airports, 11 ground crossings and four seaports.

A variety of information systems and tools used to collect and manage data on traveller health, include confirmation of yellow fever vaccination status; traveller screening; Covid-19 vaccination status confirmation during outbreaks; and cargo vaccination driver tracking. Additional information on tools can be found in Appendix 1.

Challenges

Despite the availability of several tools that support the documentation of traveller information at points of entry, several challenges have been encountered over time. This was more apparent especially during the COVID period. Some of the challenges include but not limited to:

- Multiplicity of systems with about 5 mostly bespoke systems were in use at POEs.
- Duplication of functions across the solutions; two platforms used for tracking vaccine uptake. A paper system for some vaccines, digital for others and a separate system for vaccine certificates
- System downtimes coupled up with the lack of IT support making it hard for users to capture data promptly.
- Inability to authenticate traveller information due to a lack of cross-border centralised system.
- Inaccessible records due to the use of manual tools in some cases and this led to data loss.
- Some of the data once collected is not utilised for informed decision making. This is made difficult by the unavailability of a centralised data repository.

Recommendations/ opportunities for automation

- The travel vaccination register, which is currently in paper form, can be digitised and expanded to support other vaccinations issued at the ports of entry.
- Design or adopt a solution that can conduct event-based surveillance of priority diseases at points of entry.
- Develop a comprehensive tool that consolidates similar functions e.g. can support vaccination and management of traveller documents.

2.2.2. Laboratory Based Surveillance

Laboratory based surveillance is conducted through several reference laboratory units that provide referral services linking national, international, and county laboratories. This network of laboratories is capable of performing specialised tests for priority infectious and non-communicable diseases. They conduct laboratory-based disease surveillance and provide quality assurance for the public health laboratory network.

A National laboratory data repository is currently hosted in the National Public Health Laboratory (NPHL) in Nairobi. The National Laboratory repository receives laboratory samples and surveillance data for: COVID-19: PCR, RDT; Antimicrobial Resistance (AMR); SARI/ILI FLU; TB –DR; and WBBS-SARS-COV-2.

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Information Systems in use

Labware is the most used Lab Management Information System (LMIS). Labware supports data collection and laboratory services management and is also implemented in 33 county laboratories. Health facility-based laboratories use different types of LMIS. The MOH 706A/B and MOH 207 are also used especially in the facility-based and regional laboratories in lower-level facilities. Additional information on tools can be found in *Appendix 1.*

Challenges

Despite having several systems that interface with and support the lab work, there have been several challenges experienced. These are as listed below.

- Functionalities are fragmented or duplicated across systems.
- Universal patient identifier is not fully implemented to enable patient tracking across systems & health processes.
- Multiple systems store data with no proper mechanism for access and use.
- With the frequent transfers of healthcare providers, there is lack of proper training guides for the proper uptake and use of the systems.
- Outdated system functionalities making it difficult for integration and use.
- Lack of a standardized method of lab requisition services i.e., for equipment calibration services.
- Lack of a standardized process of acquiring authorization to access laboratory systems.

Recommendations/opportunities for automation

- Digitise the Case investigation lab order-Results cycle to strengthen data consistency and integrity.
- Adopt a Unique Person Identification system for Lab Epi interface.
- Link LIS in facility-based labs with the regional Public Health Laboratory networks.
- Strengthen and digitize routine laboratory reports from facility-based labs through LIS/EMRs.2.2.2.

2.2.3. Immunisation in Kenya

Kenya provides several antigens for preventing priority diseases. Most immunisation programs are designed for children in line with Kenya's routine immunisation schedule and recommended vaccination for special risk groups, for immunocompromised and special risk groups in line with Kenya Immunisation policy guidelines, 2023 Some selected sub-populations, such as food handlers who are required to receive periodical typhoid vaccines, and travellers to selected countries that require yellow fever vaccines, paper-based and digital tools are used to record, collate, summarise, and report routine immunisation data and adverse events following immunisation. During Covid-19, to support the monitoring of vaccine uptake, Chanjo.ke was developed and adopted. Additional information on tools can be found in *Appendix 1.*

Challenges

- The system has no unique identification, especially for children.
- The lack of a centralised data repository makes it difficult for key stakeholders to access and use data effectively.
- Stock-outs are a consistent challenge.
- There are data quality issues due to manual collection processes.

Recommendations/opportunities for automation

To support the ongoing efforts of ensuring effective and efficient service delivery of the unit, the following are the key recommendations for consideration.

- Adoption of EHR/EMRs for recording vaccine administration and reporting uptake of vaccinations across various programs and for vaccine commodity management across all health facilities.
- Adoption of a Vaccine Information Management System for all antigens issued to different sub-populations.

2.2.4. Vector-Borne and Neglected Tropical Diseases

Neglected tropical diseases (NTDs) are a group of chronic disabling infections affecting more than one billion people worldwide, mainly in Africa and mostly those living in remote rural areas, urban slums, or conflict zones. The NTDs range from Trypanosomal parasites (Leishmaniasis, Human African Trypanosomiasis, Chagas disease/American trypanosomiasis); Helminth parasites (Schistosomiasis, Lymphatic Filariasis, Onchocerciasis, Drancunculiasis, Echinococcosis, Taeniasis/Cysticercosis, Ascariasis, Trichuriasis, Hookworm, Strongyloidiasis); Bacterial infections (Leprosy, Buruli ulcer, Noma, Trachoma, Yaws) and Viral infections (Japanese Encephalitis, Rabies) Strategic pillars for successful control of NTDs include: Acceleration of programmatic action; intensification of cross-cutting approaches; enhancing resource mobilization; advocacy, and sustainability; strengthening leadership capacity; communication; and collaboration for the elimination of NTD. [11]

Challenges

- Surveillance of VBDs and NTDs is not mainstreamed into the health system.
- The unit currently utilizes WHO data collection tools with no customised MOH data collection and reporting tools.
- There are no clinical diagnostic tools making it difficult to detect and respond to NTDs on time.

Recommendations/opportunities for automation

- Developing custom MOH tools for data collection and reporting.
- Incorporating surveillance of VBDs and NTDs into the mainstream health system.

2.2.5. Passive Surveillance for ILI and SARI

In Kenya, influenza surveillance is conducted through syndromic surveillance commonly referred to as the influenza-like Illness (ILI) and Severe Acute Respiratory Infection (SARI) surveillance system. This ILI/SARI surveillance is achieved through a sentinel surveillance approach. A total of 10 sentinel surveillance sites distributed across the country routinely collect ILI and SARI surveillance data and submit it for use in influenza surveillance.

Currently, manual clinical care processes are used to collect symptoms and diagnose respiratory diseases. Surveillance officers stationed at the sentinel sites routinely abstract data from these clinical records and registers through a process that is parallel to the clinical care one. The collected data is recorded into Netbooks that are used to report cases of ILI/SARI. This separation of clinical care processes from the disease surveillance process has resulted in inefficiency as it involves double data entry. The burden of care provision and case detection lies with the health care providers who are responsible for documenting case details that are later abstracted by surveillance officers. This manual process of surveillance data collation and reporting is inefficient, prone to errors and under-reporting of cases.

Despite the existence of EMRs, currently, none of the sentinel facilities have a fully automated system that could flag possible cases despite there being some systems in use at the OPD and in some cases IPD. Additionally, disease surveillance is not integrated within the clinical care workflow in the facilities. It is thus difficult to routinely get insights on total clients screened compared to those who screened positive for diseases under surveillance. Integrating routine, the disease surveillance process within routine clinical care could simplify case detection, data collection and reporting. This process could be enhanced by leveraging on digital health solutions used in public health facilities to set up an integrated passive surveillance system that supports disease surveillance activities broadly.

Recommendations/opportunities for automation

- Digitisation of passive surveillance through routine Point of Care clinical care solutions.
- Adoption of passive surveillance for respiratory illness and other priority disease conditions.

2.2.6. Disease Surveillance through Strategic Health Programs.

The Ministry of Health has designated Divisions of National AIDS and STI Control Program (NASCOP), National Tuberculosis and Leprosy Control Program (NLTB) and the Division of Malaria Control (DOMC) to oversee implementation of these three disease conditions. Although the three divisions contribute aggregate data to KHIS for Indicator Based Surveillance, they also conduct additional disease surveillance activities.

The Division of Tuberculosis Program relies on TIBU for TB surveillance: Through a network of County-based TB and Leprosy Coordinators (CLTC) for surveillance of New TB cases, Active cases of Tuberculosis and Drug-resistant TB among others are recorded and reported to the TIBU platform. Data on TB cases and TB-HIV co-infection is routinely abstracted from routine clinical files and MOH registers and entered by County and Sub-County (S-CLTB and CLTB) coordinators into TIBU, a digital health solution and uploaded to a central database from where it is analysed to monitor TB incidence, prevalence and outcomes.

The Division of National AIDS, Sexually Transmitted Infections Control Program (NASCOP) implements HIV Case Surveillance and HIV Case Based Surveillance. Data is collected through EMRs and analysed at national level to inform public health response interventions. These routine surveillance activities, including drug resistance and mortality surveillance is complemented by several of periodic surveys e.g. Kenya National Public Health Impact Assessments among all populations and Integrated Bio-behavioural Surveys among Key Populations.

The National Malaria Control program conducts routine surveillance of uncomplicated and complicated surveillance. They monitor malaria prevention interventions, the incidence, treatment and outcomes of malaria cases in malaria endemic areas, seasonal malaria transmission zones and malaria epidemic prone through outpatient and in-patient registers. Malaria surveillance in malaria epidemic prone areas takes the sentinel surveillance approach. Sentinel surveillance involves routine collection of data on malaria cases, treatment and outcomes selected facilities within Malaria outbreak prone areas. The collected data is analysed against pre-set thresholds and act as signals for potential outbreaks and guide malaria control interventions to avert potentially devastating outbreaks.

2.3. One Health Approach

One Health (OH) promotes the decompartmentalization of human, animal, and ecosystem health. It acknowledges the relationship between people, animals, plants, and environment and requires that traditional boundaries between disciplines and sectors are transgressed and that all relevant stakeholders are involved in the definition and management of health problems. As human populations expand, close contact is created between humans and animals. The result is the risk of diseases being shared between the two through the contamination of water and food supplies by diseased food animals as well as recreational activities that involve humans and animals.

Kenya's One Health Strategic Plan for Prevention and Control of Zoonotic Diseases (2021–2025) prioritizes strengthening multi-sectoral collaboration and surveillance across the human, livestock, wildlife, environmental, and food safety domains. The plan recognizes that robust prevention and control of zoonotic and emerging threats require joint surveillance systems, data sharing, and coordinated response mechanisms. In line with the 2024 Joint External Evaluation (JEE) recommendations on One Health, Kenya is encouraged to build on this foundation to formally include the Ministry of Environment, Climate Change and Forestry, and other relevant sectors. The framework recommends developing and operationalizing a national interoperability platform to integrate surveillance systems across sectors and establishing an overarching national One Health policy to provide legal and institutional anchorage for multisectoral collaboration among other related recommendations. By embedding these actions into the ongoing One Health agenda, Kenya will advance its progress toward achieving a fully integrated, multisectoral surveillance system that aligns with both the One Health Strategic Plan (2021–2025) and KNPHI. The ZDU plays a critical role in coordinating surveillance of priority zoonotic diseases through the county one health units. Where there is no COHUs, the use of Kenya animal bio surveillance (KABS) and KHIS suffices. It highlights the sub-optimal involvement of environmental health as a major gap in One Health.

The Strategy also highlights the gap in the adoption of ICT, especially in livestock and wildlife surveillance and notes it as an intervention area that will strengthen One Health. However, there is a critical gap in data sharing as the two systems of reporting are not inter operable.

There are three reporting tracks under One Health. Under IDSR which is extensively described in Chapter 3; priority zoonotic diseases are under surveillance through surveillance officers who collect data using standard reporting tools and report to KHIS (DHIS2).

2.4. Disease Surveillance in Domestic and Wild Animals

The vision of Kenya's Zoonotic Disease Unit (ZDU) is the elimination of zoonotic disease threats to ensure well-being of humans, animals and the environment in the country. The goal is to reduce the burden of zoonotic diseases through a One Health approach. The departments within the ZDU look at the unique interactions between animal and human health under the One Health approach. Although the legal framework is weak, Kenya's Health Act (2017) and proposed legislation support this approach.

Information Systems/ tools in Use

Reporting of livestock diseases: Livestock diseases are reported by sub-county veterinary officers to the epidemiology and Economic section under director of veterinary services.

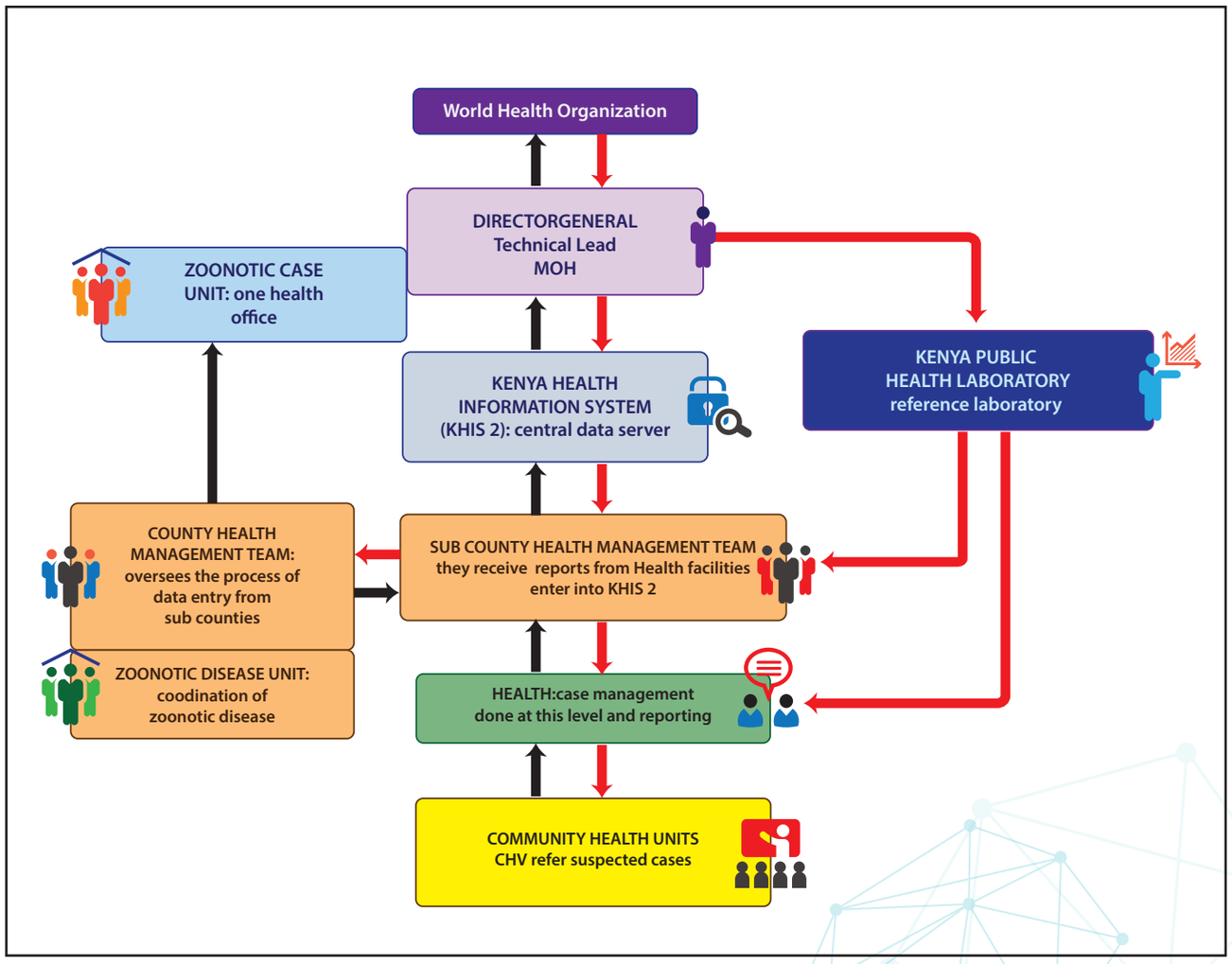


Figure 5: Interaction Between Human and Animal Data

The Ministry of Health and the Ministry of Livestock collect data from meat inspections in slaughterhouses, reported through county District Veterinary Officers (DVOs), for surveillance purposes. The Ministry of Wildlife conducts opportunistic surveillance of morbidity and mortality in wildlife, compiles the data, and reports it to the Kenya Livestock and Wildlife Syndromic Surveillance System, sometimes via mobile phone technology (MPT) through an app known as the Kenya Animal Bio Surveillance (KABS) tool, currently a pilot.

Animal health reports are routinely submitted to Office International des Epizooties (OIE) via the World Animal Health Information System (WAHIS) and to African Union Inter-African Bureau of Animal Resources (AU-IBAR) via the Animal Resource Information System (ARIS). AU-IBAR supports countries to establish data management and sharing platforms.[13]

Animal disease collection tools

The following tools are used to collect data on animal diseases:

- Sanitary Report Form: This form is filled by field surveillance officers, private practitioners, meat inspectors, etc. to report animal diseases and events in real time.
- Zero Report Form: It captures the absence of priority animal diseases every time a surveillance officer examines an animal.
- Community Diseases Report Form: This form is used by community disease reporters to report observed clinical signs on real time basis.
- Sample submission form: This form is used by practitioners (public and private) who submit samples to the lab.
- Livestock vaccinations form: It captures vaccination figures for all livestock diseases as they are undertaken at the ward level.
- Event-based surveillance for picking predetermined signals for upward verification and response by the surveillance officers. Jointly implemented with the MOH team.

Reporting Tools

The four reporting tools used are:

- Kenya Animal Bio-Surveillance System (KABS) [14]
- District Health Information System 2 (DHIS2);
- Animal Resource Information System (ARIS);
- World Animal Health Information System (WAHIS).

Details on these tools and how they are applied can be found in Appendix 1

Human Surveillance Tools

These include:

- The Rift Valley Fever ESS tool.
- Active outbreak investigation tools, such as the Case Investigation Form.
- Vaccine use and monitoring forms.

Challenges:

- Weak reporting structures for reporting animal diseases; several data collection and reporting tools are currently not digitised.
- Weak sample referral system especially for zoonotic diseases across the board.
- Weak linkage with Anti-Microbial Resistance initiatives.
- Lack of interoperable data systems.
- Inadequate information sharing between surveillance and research institutions.

The Ministry of Health established the Zoonotic Disease Unit to ensure seamless linkage of IDSR with environmental and animal health (livestock and Wildlife) surveillance and response initiatives. As such, the challenges outlined above can be addressed through implementation of the One Health Approach summarized in section 2.3 above

2.5. Environmental Health Surveillance

The Water, Sanitation and Hygiene (WASH) unit is under the Division of Environmental health in the Department of Public Health. WASH is mandated to ensure easy access to water, hygiene, and proper sanitation in the community. The unit's focus points are water, sanitation, and hygiene in healthcare facilities, in schools, and in communities (households), as well as Environmental Surveillance. Within environmental surveillance, the main goal is to mitigate disease outbreaks caused by environmental health factors. Communities are monitored for open disposal of human waste and supported in becoming Open Defecation Free (ODF). Community Led Total Sanitation (CLTS) has five stages: triggering; follow-up; verification; certification; and quality assurance that must be completed for a village to be declared ODF. Reporting was mostly manual and time-consuming. To resolve this, MOH introduced the Community-Led Total Sanitation (CLTS) Real Time-Monitoring Information System to monitor households and community engagements. The real time monitoring system has a dashboard containing data from the community, county, and national level. The data that feeds the dashboard is collected through a mobile application (Kobo Collect).

Those communities that have moved beyond the ODF are introduced to Joint Monitoring Program (JMP) parameters. The WHO and United Nation Children's Fund (UNICEF) Joint Monitoring Programme for Water Supply, Sanitation and Hygiene (JMP) has reported country, regional and global estimates of progress on drinking water, sanitation, and hygiene (WASH) since 1990. A JMP parameter has been used to rank community WASH programs into three colour-coded grades:

- Grade 1: Limited open defecation free community.
- Grade 2: Improved; safe and sustainable sanitation.
- Grade 3: Safely managed sanitation: clean and healthy household.

In 2022, the Real Time Monitoring Information System expanded to cover not only communities but also households and institutions.

Despite the government's commitment to achieve Sustainable Development Goal (SDG) target 6.2 by 2030, there is no national monitoring system for safely managed sanitation and hygiene services. The CLTS Rural-Urban Sanitation Hygiene Real-Time Monitoring Information System (RUSH RT-MIS) provides information on rural sanitation with an emphasis on ODF status, with no attention paid to excreta containment, emptying, transport, and treatment. Additionally, CLTS did not address the implementation of market-based sanitation, and it did not cover the entire country. These are the main factors that made the transition necessary.

RUSH RT-MIS focuses on eight key areas of interest: Water sanitation; water supply; hand hygiene; contamination; waste management; lighting; climate resilience; and environmental manipulation. The system is expected to be launched in 2024.

At the various places several key factors are checked. The idea is to ensure that these facilities are accessible and usable for instance by Persons with Disabilities (PWD).

- At the household level, matters of sanitation are checked: type of toilet, hand washing facilities, waste disposal, environmental cleanliness, waste management.
- At the community level, structures (markets, public toilets, and access to water) are examined.
- Within the institutions, access to water, handwashing facilities, waste disposal and management, menstrual hygiene, and types of toilets are checked.
- At health facilities, the focus is the safety of the patient and caregiver (not getting hospital-acquired infections), type of toilets, handwashing stations, and waste management.

Challenges

- Currently, the WASH assessment tool, MOH 708, only captures two indicators: number of toilets, and number of hand-washing stations per village.
- IDSR datasets are not adequately linked to existing environmental monitoring datasets. To create an early warning system for outbreaks, it is necessary to enhance the current data.

- There's no linkage on environmental surveillance interventions between the Ministry of Health and the Ministry of Environment.

Recommendations/ Opportunities for Automation

- Revise MOH 708 to capture additional information beyond communities, and have it integrated into KHIS, this would ensure swift and easy access to data as well as limit opportunities for loss of data.
- Linkage of environmental surveillance datasets with IDSR is weak enriching existing data to establish an early warning system for outbreaks.

Broader environmental surveillance efforts are conducted in Kenya through the Ministry of Environment, as such environmental health surveillance data is not fully integrated with IDSR and Animal disease surveillance efforts as envisioned in the One Health Strategic Plan. There are significant efforts to support this, especially with environment component now embedded under ZDU within NPHI.

2.6. Food Safety Surveillance

The governance of food safety in Kenya is grounded in the constitutional framework of devolution (Section 4 of the Constitution of Kenya), which places service delivery, including preventive and promotive health, under county governments. At the national level, the Division of Food Safety within the Ministry of Health (MOH) provides policy direction, issues circulars, and maintains liaison with the Council of Governors (CoG). Communications from the national government are cascaded through the CoG to counties, where County Public Health Officers (PHOs) operationalize the directives. Sub-county PHOs, as the frontline actors, are specifically tasked with food safety functions including market surveillance, food sampling, and enforcement of food premises regulations.

Currently, food safety surveillance faces several structural and system-related challenges. Market surveillance tools are not standardized across counties and remain largely manual. Sampling forms accompanying specimens sent to the National Public Health Laboratory (NPHL) or mini-labs are also paper-based, making it difficult to integrate into the national e-disease surveillance roadmap. While the MOH 708 tool embedded in the Kenya Health Information System (KHIS) represents a positive step toward digitization, it is incomplete—several food sampling components are missing. An ongoing review of MOH 708 aims to address these gaps to ensure interoperability and alignment with the broader national digital health architecture.

Progress is, however, being made through the development of an integrated food safety surveillance system, spearheaded in collaboration with the KNPHI. This initiative seeks to standardize tools, digitize data capture, and harmonize food safety information flows across national, county, and sub-county levels. The system is designed to be interoperable with KHIS and to align fully with the national e-disease surveillance roadmap. Once operational, the platform will support real-time data entry by PHOs, automate sample tracking from collection to testing, and strengthen the feedback loop between counties, NPHL, and the national Division of Food Safety. This represents a major step towards embedding food safety surveillance within the broader KNPHI mandate of coordinating public health intelligence in Kenya.

Sample testing at mini labs is carried out by trained PHOs rather than laboratory technologists, reflecting the preventive and promotive health orientation of counties. Confirmatory tests, however, require referral to the NPHL, with the County PHO serving as the focal point for coordination. Proposals are underway to establish regional laboratories to decentralize confirmatory testing, reduce turnaround time, and enhance responsiveness. Digital platforms for relaying laboratory results from the NPHL or future regional hubs to counties will further strengthen the surveillance-response cycle.

The current system of reporting is laborious. Food safety focal points (PHOs) notify the national Division of Food Safety of contaminated products through email in compliance with existing governance framework both at county and national levels, after which circulars are issued to all 47 counties if widespread contamination is confirmed. The lengthy approval chain delays timely action. Digitizing this notification and escalation process within the integrated food safety surveillance system and the national e-disease surveillance platform will allow for real-time alerts and faster containment of risks.

In practice, food sampling is triggered by routine surveillance, directives issued through circulars, or complaints from communities. Owner consent is sought before destruction of food deemed unsafe; where consent is withheld, the courts are engaged for authorization. This legal framework ensures due process, while allowing for the protection of public health.

In addition to sampling, sub-county PHOs trained in food safety and quality assurance, conduct risk-based food inspections, monitor food fortification compliance, enforce food premises guidelines, and oversee food handler screening and hygiene practices. Surveillance data on the "top ten most consumed foods" in a locality guides risk categorization and prioritization of sampling. However, because food safety at county and sub-county levels remains embedded in preventive and promotive health departments, unlike at national level where it operates as a standalone division, any future interoperability framework between counties and the national government must account for this structural difference.

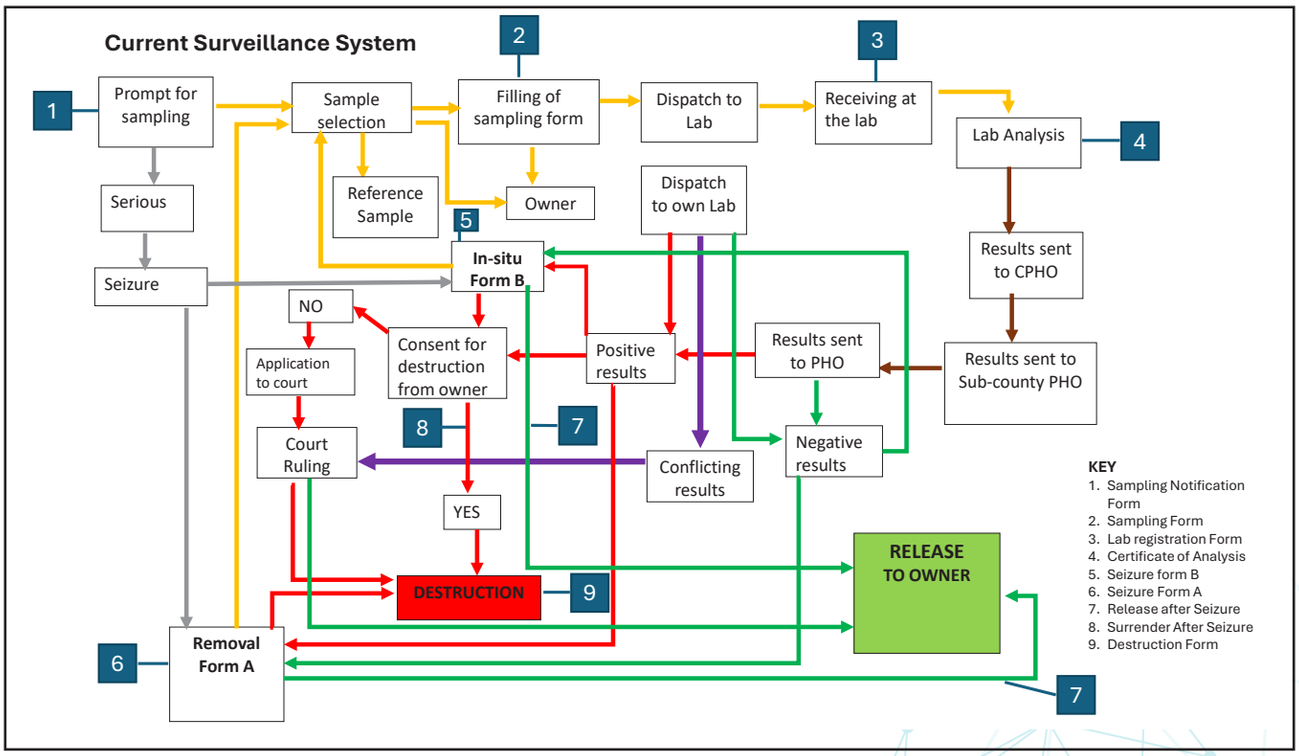


Figure 6: Food Sampling Process Flow

Challenges

- **Sample Management:** Following sample collection, the processing forms are filled out by hand. Results are also processed manually. It is challenging to track samples using this procedure, which is prone to errors and delays.
- **Reporting challenges:** It is difficult to obtain a comprehensive report of all food samples and tests conducted in a county over a period since 2013 when food sampling was devolved to Counties.

Recommendations/ Opportunities for Automation

- Standardization and digitization of market surveillance and sampling tools..
- Integration of food safety modules within KHIS (review of MOH 708).
- Establishment of an integrated food safety surveillance system under KNPHI to align with the national e-disease surveillance roadmap.
- Decentralization of confirmatory testing through regional laboratories.
- Development of a digital results-relay and notification platform.
- Recognition of the pivotal role of sub-county PHOs as the frontline implementers of food safety mandates.

2.7. Public Health Emergency Operation Center (PHEOC)

A Public Health Emergency Operation Center (PHEOC) is a physical location that serves as the central platform for coordination of response to public health emergencies including international threats from emerging and re-emerging epidemics/pandemics, as mandated by (IHR) 2005. [15]

The PHEOCs promote collaboration among response agencies to ensure a goal-oriented and effective response to public health emergencies. PHEOCs play a critical role in managing and responding to health crises ensuring a coordinated and efficient approach to safeguarding public health.

The Kenya National Public Health Emergency Operations Center (KPHEOC) was inaugurated in May 2016. It is situated within the Kenyatta National Hospital grounds. The establishment of county PHEOCs is ongoing. As of April 2024, cumulatively 22 county PHEOCs have been established. PHEOCs operate within the WHO framework for PHEOC. [15] The PHEOC operations are guided by Kenya's national PHEOC SOPs, handbook, [16] and strategic plan 2023-2027. [17] The key components of PHEOC include policies, plans, and procedures; trained and skilled staff; physical infrastructure; information systems and standards, and ICT infrastructure. The National PHEOC space in Kenya was not purpose-built but existing structures were repurposed to host PHEOC. The PHEOC space should be in a secure and accessible location with sufficient space to accommodate a call center, PHEOC manager's office, watcher's workstation, meeting boardroom, reception, washrooms, and kitchen.

The PHEOC should have at least five (5) core staff in line with its core functions of management, operations, planning, logistics finance, and administration. In response mode, an Incident Management System (IMS) is constituted. The IMS is domiciled within the organizational functions of the PHEOC as mentioned above. Surge staff and subject matter experts are deployed to the PHEOC during response mode depending on the size and complexity of the public health emergency.

PHEOCs have five main functions:

- Coordination: PHEOC's coordinate information flow, resource allocation, and response efforts among stakeholders.
- Situation Monitoring: They continuously monitor the situation, collect data, and analyze trends to provide situational awareness.
- Resource Management: PHEOC's manage resources such as medical supplies, personnel, and logistics.
- Communication: The PHEOC facilitates communication between response agencies, government bodies, and the public via briefs, spot reports, situation reports and media briefings. The PHEOC also operates a 24-hour call center where the public can report public health events through toll-free lines. An information management system is required for the timely generation of PHEOC reports.

- Decision-Making: PHEOC's support decision-making by providing real-time information to incident commanders.

Modes of Operation of the PHEOC

The PHEOC operates at different levels referred to as modes, depending on the magnitude and risk of a prevailing public health threat. Different activities are undertaken during the various modes of operations.

1. **Watch Mode**

In the watch mode, routine day-to-day operations are undertaken by watch staff. The watchers monitor disease trends through periodic incident-based surveillance (IBS) reports. They also track events that may represent threats to public health through hotlines, media scanning, and community and health facility event reporting.

2. **Alert Mode**

The alert mode is the standby phase of activation when an emergency event is imminent or has occurred. The PHEOC conducts intensive monitoring of an incident or event in preparation for a potential PHEOC activation. In this mode, watch staff continue with their responsibilities while the operations personnel are assigned additional responsibilities i.e. preliminary risk assessment, alert notifications, activation of plans and deployment of the rapid response teams (RRT).

3. **Response Mode**

During the response mode, the PHEOC is partially or fully activated guided by pre-defined criteria for levels of activation corresponding to levels of response. The lowest level of response addresses relatively lower scale events for which all response activities are largely within the capabilities and resources of the PHEOC, and low-level augmentation is required.

Activation of the PHEOC

The Director General (DG) of Health (at the county level the County Director of Health, CDH or the County Director responsible for Public Health) gives directives for activation of the PHEOC following a proposal by the PHEOC manager. The Cabinet Secretary (CS) or other responsible authority may also directly provide directives for activation if needed.

Deactivation of the PHEOC

When the response is declared over, the PHEOC will be deactivated and return to routine monitoring. This follows the de-escalation of PHEOC activities. De-activation plans entail:

- Orderly and progressive cessation of activities and functions.
- Return of resources to their original point.
- IMS functions beginning deactivation processes.

PHEOC Milestones

Since its establishment, the KPHEOC has coordinated response for COVID-19, circulating vaccine-derived poliovirus 2, yellow fever, measles, Rift Valley Fever (RVF), leishmaniasis, anthrax, cholera, El-Nino and the Embakasi fire incident. The KPHEOC has also conducted an intra-action review of COVID-19 and an after-action review of cholera. Other milestones include the development of an online public health emergency management (PHEM) curriculum, intermediate PHEM training of national and county officers, rapid response training of national and county officers, and simulation exercises on RVF and Ebola Virus Disease (EVD).

Currently, there is lack of a publicly available platform mechanism to promote data sharing between the various stakeholders involved in public health emergency preparedness and response. There is a limited amount of information on the equipment available in the established PHEOCs and most PHEOCs were yet to optimally adopt ICT for their operations. Additionally, different reporting formats used for reports submission to the national PHEOC by counties, making data analysis a challenge.

2.8. Summary of key issues and justification for an e-surveillance strategy roadmap.

2.8.1. Coordinated Digitisation Efforts

Currently, no unique patient identifier exists to track patients across all systems and processes. Adopting Unique Person Identifiers would resolve this by providing a platform that facilitates unique identification of clients, sample management, contact tracing and deduplication of patients who access services in more than one facility.

A review of surveillance systems indicates several key cross-country challenges. A common theme is the prevalence of manual, paper-based surveillance, and hybrid surveillance approaches where both manual paper and digital systems interact to conduct surveillance. This can be resolved through efforts and investments towards progressive digitisation.

There are multiple, duplicative, and in some cases bespoke data collection systems supported by different divisions even where singular solutions can be implemented to enhance diseases surveillance. This underscores the need for a coordinated approach to digitisation that draws from the overall MOH digitisation efforts. There is an urgent need to clean up obsolete applications and transition data resident in them to modern solutions that future-proof the directorate's digitisation by basing them on ongoing sector-wide digitisation and a transformation of the digital landscape initiatives.

There are several challenges facing system capabilities: Firstly, functions are currently duplicated and fragmented across various solutions and systems. Secondly, security, standard compliance and user experience are all subpar. A major challenge is that features are outdated, and no one is available to update software to the most recent versions. The Ministry of Health also offers sub-optimal IT support; there is a lack of dedicated IT staff available for quick assistance. Some applications frequently experience system outages, making it difficult for users to quickly collect data. To resolve this, it is imperative that MOH become the custodian of all applications and maximally support their re-use.

Sub-optimal availability and up-time of the surveillance systems was prevalent across the board. This is complicated by heterogenous hosting plans and databanks where surveillance datasets are held. This situation can be resolved by applying national standards that guide where data is hosted.

The recently passed Digital Health Regulations contain provisions on health data governance. Datasets collected through the digital health solution described in section 3 above were not well documented. The data dictionaries for most disease surveillance solutions are not standardized or mapped with KNHTS. The COVID-19 pandemic exposed the imbalance between the imperative to protect health data (including sensitive personal data), the need to ensure equity in data use and the requirement to maximise the health value of data as enshrined in the principles of health data governance. There digital health directorate should implement proper digital health and data governance structures in line with recently enacted legislation and policies.

There main challenge facing health information exchange is that there is no standard concept management. Most systems do not have data dictionaries outlining the data being collected, making it difficult to exchange data across systems. The COVID-19 pandemic exposed the imbalance between the imperative to protect health data, including sensitive personal data, and the need for equity and maximized health value of data not based on persons. Proper governance structures would resolve HIE challenges by standardising data exchange and balancing health data protection with equity and maximised data.

It has been difficult to store, share and use data for various reasons: Manual records are inaccessible, leading to data loss. Furthermore, even where data has been gathered, little analysis has been done on it. The result is an inability to draw conclusions from it to make well-informed decisions. Thirdly, data quality issues arise when data is collected manually. Fourth, data is not visible to all the required stakeholders. Lastly, data is isolated within certain systems. The remedy for this would be the Ministry of Health taking custody of all surveillance data and containing it within a centralised repository complete with an analytics platform that can host standard outputs and support self- service data use as much as possible. Supporting standard data sharing agreements would ease data accessibility, use, and sustainability.

User capacity needs strengthening. End users currently are inadequately trained. Learning materials are limited, resulting in insufficient knowledge of system use. The Ministry of Health can work with Partners to host easily accessible building materials in the MOH Virtual academy.

2.9. Linkage with sector-wide digitisation efforts

The Ministry of Health established a Health Information Exchange (HIE) within its proposed Digital Superhighway. The HIE has the potential to create new opportunities for broadening the number of facilities reporting as part of public health surveillance thereby expanding the coverage of disease surveillance initiatives.

The digital health certification framework, to be enforced as part of the HIE will ensure all implemented digital health solutions are compliant with disease surveillance requirements. The digital health superhighway offers services that act as incentives for the private sector to share data with the Ministry of Health, this will further bolster disease surveillance efforts. It also offers opportunities for Automated Indicator Reporting (AIR). AIR involves automated generation and reporting of service statistics/Indicators to the Kenya Health Information System (KHIS). This approach will strengthen Indicator Based Surveillance by eliminating data entry inefficiencies, ensuring Indicator Based Surveillance (IBS) statistics can be traced to cases and support EMR enabled passive disease surveillance across public and private sectors.

Shared digital utilities within the digital superhighway offer an opportunity for disease surveillance programs to strengthen long-standing challenges in communicable disease surveillance efforts. The key components are as outlines below.

- The Master Facility/Community Registry developed by the Kenya Ministry of Health serves as an index to all health facilities in the country. This will be used to locate and identify facilities and surveillance sites.
- The Kenya National Health Terminology Service (KNHTS) will ensure the standardisation of concepts, terms and case definitions across systems and ensure seamless standards-based semantic interoperability.
- The adoption of a Client Registry (CR) for Person Identity Management will strengthen quality of surveillance data and ensure Unique Person Identification is practiced in counting cases during epidemics reported across facilities and ensure epidemiological data links seamlessly with laboratory diagnostics at the epi-lab interface. This will further strengthen case-based surveillance efforts.
- The proposed national Health Data bank will contain minimum Shared Health Record (SHR) for everyone. Similarly, health records may be kept in County Health data banks as designated for monitoring and evaluation, or as part of a regional EMR system.
- Various POC systems may exist within a healthcare facility or communities including e-CHIS, Electronic Health Records, Pharmacy and Lab information. These systems may be linked together to improve patient care, monitoring, and reporting.
- The National Enterprise Service Bus connects different systems together nationwide. The E-Surveillance roadmap assumes that the MOH will implement Service Oriented Architecture (SOA), allowing systems to access services and other systems without having to connect directly to each one. Instead, systems and services will be connected by means of an Enterprise Service Bus (ESB) that can route messages to the appropriate services. This allows the overall system to grow, change and adjust as necessary. It means that every system does not need to know where every other system is located, or specific technologies implemented.

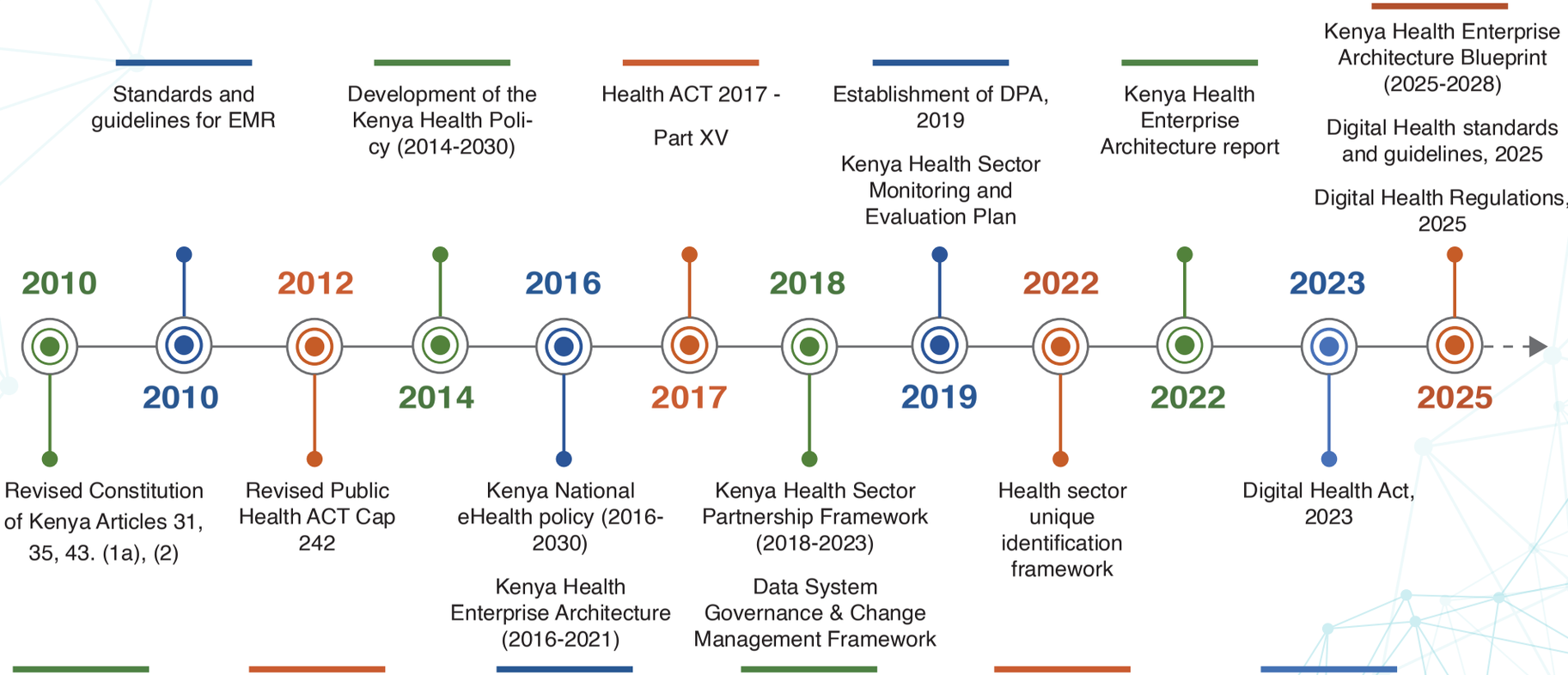


Figure 7: Digital Health and Data Governance Legislative Timeline

2.9.1. Digital Health and Data Governance: Legislative Timeline and Landscape

The Government of Kenya has drafted and passed several laws and policies to drive digital transformation of health sector. The Health Act, 2017, mandates MOH to set up a Comprehensive, Integrated Health Management Information System. The Data Protection Act, 2019 (DPA), focuses on the protection of data subjects. The Digital Health Act, 2023 (DHA) builds on the DPA, contextualising it to health. The DHA establishes new governance structures. DHA is the legal basis for the Health Data Governance Framework (HDGF). DHA regulations operationalise the governance structures in HDGF and adopt templates and SOPs within it for Health Data Governance. DHA and HDGF apply the Global Principles of Health Data Governance that ensure data protection is balanced with the need to maximise the health value of data through data sharing and ensure equity in health data collection and use. These principles are especially critical for disease surveillance programs that heavily generate and use data for decision making.

3

Future State: Achieving an Electronic Integrated Disease Surveillance & Response Ecosystem.



3.1. Aim

The Kenya National Public Health Institute (KNPHI) cases manual data collection systems for monitoring of priority diseases and public health events. The Ministry of Health recognises that inefficiencies associated with manual paper surveillance systems can be overcome by transitioning to electronic platforms for disease surveillance.

The goal of KNPHI is to establish an integrated e-Surveillance system that supports public health activities broadly, accurately and in near real-time. This e-surveillance platform will be designed to facilitate and enable the One Health approach for surveillance of public health threats by all actors.

3.1.1. Objectives

- The specific objectives of the establishment of e-surveillance are outlined below.
- To develop a multichannel e-surveillance system interoperable with other surveillance systems to provide early detection, monitoring, reporting, and response to disease outbreaks (event-based surveillance) for timely public health intervention.
- To link primary surveillance data sources (case reports, syndromic surveillance, lab reports, and mortality reports) to e-surveillance systems.
- To develop a reporting and analytics layer on seroprevalence, outbreak investigation indicators, and predefined transmission dynamics.
- To provision an Early Warning System that calibrates Watch-Warn-Response metrics on disease vectors and provides Command and Control visibility and alerts to authorised personnel.
- To coordinate the network of surveillance partner processes and systems within MOH surveillance ecosystem to support interoperability and data exchanges.
- To efficiently offer disease surveillance and outbreak response services by KNPHI to the public.

3.2. Thematic Areas, Strategic Focus Areas and Activities Towards Realisation of Electronic Surveillance (e-Surveillance)

This surveillance roadmap will be based on strategic focus areas under the following seven (7) thematic areas.

Thematic Area	Description
Thematic Area 1	Develop a National Disease Surveillance Repository at the Ministry of Health to support data driven decision making.
Thematic Area 2	Strengthen Routine Surveillance through transmission and hosting of disease surveillance data in a national disease surveillance repository.
Thematic Area 3	Revitalise Lab-Epi Interface through automation of lab sample and results management.
Thematic Area 4	Strengthen Outbreak Response capability by digitising outbreak response workflows.
Thematic Area 5	Strengthen Data Governance Structures and facilitate Data Sharing and use for public health surveillance.
Thematic Area 6	Support the deployment of the necessary infrastructure (hardware and software) needs for operationalizing a National Disease Surveillance Repository.
Thematic Area 7	Develop a Workforce for effective routine e-surveillance and outbreak response.
Thematic Area 8	Digitise Operations of Public Health Emergency Operation Centers for effective surveillance, pandemic preparedness, and response

Table 2: Thematic Areas towards Realisation of electronic Surveillance

3.2.1 Thematic Area 1: Develop a National Disease Surveillance Repository at the Ministry of Health to support data driven decision making.

3.2.1.1. Focus Area 1.1: Development of National Disease Surveillance Repository and Visualizations

The Ministry will establish a national disease surveillance repository; the repository will serve as a one-stop shop for data on all diseases under surveillance and data from all surveillance initiatives. The national repository will ingest data from both new and existing routine surveillance, laboratory surveillance solutions and outbreak response initiatives in Kenya. The National Disease Surveillance Repository will be hosted within the MOH computing infrastructure and will facilitate ownership and integration of all surveillance datasets in one place at the national level. The repository will be designed and built in line with MOH approved standards. The repository will acquire data from electronic data sources and consist of a staging area, a database, integrations, and an analytics platform. The components of the platform will be as shown below.

- **Data Linkage Module:** This is centralised repository to support data storage and analysis of all surveillance data captured by various clinical, lab and aggregate systems supported by MOH and its partners. The repository is expected once its functional to be hosted with the Ministry of Health Kenya. Figure 7 below describes the proposed architecture of the National Disease Surveillance Repository.

- **Predictive Analytic layer:** The predictive analytic layer will include a Machine Learning model within the repository trained using data from all sources including ECHIS, mDharura, KHIS etc. The predictive analytic layer will be set and configured to predict disease outbreaks using EBS early warning signals.

- **Disease Surveillance Analytics Layer:** The Analytic Layer of the National Disease Surveillance Repository will consist of a reporting layer, that serves business intelligence tools which will display disease surveillance be analytic dashboards that will present key performance indices (KPIs) needed for monitoring disease patterns by surveillance and disease outbreak response teams. The surveillance dashboards will be targeted at actors at different levels of the disease surveillance ecosystem who will be provided with secure access to information when the need arises.

Architecture of The National Disease Surveillance Repository

NATIONAL DISEASE SURVEILLANCE REPOSITORY ARCHITECTURE

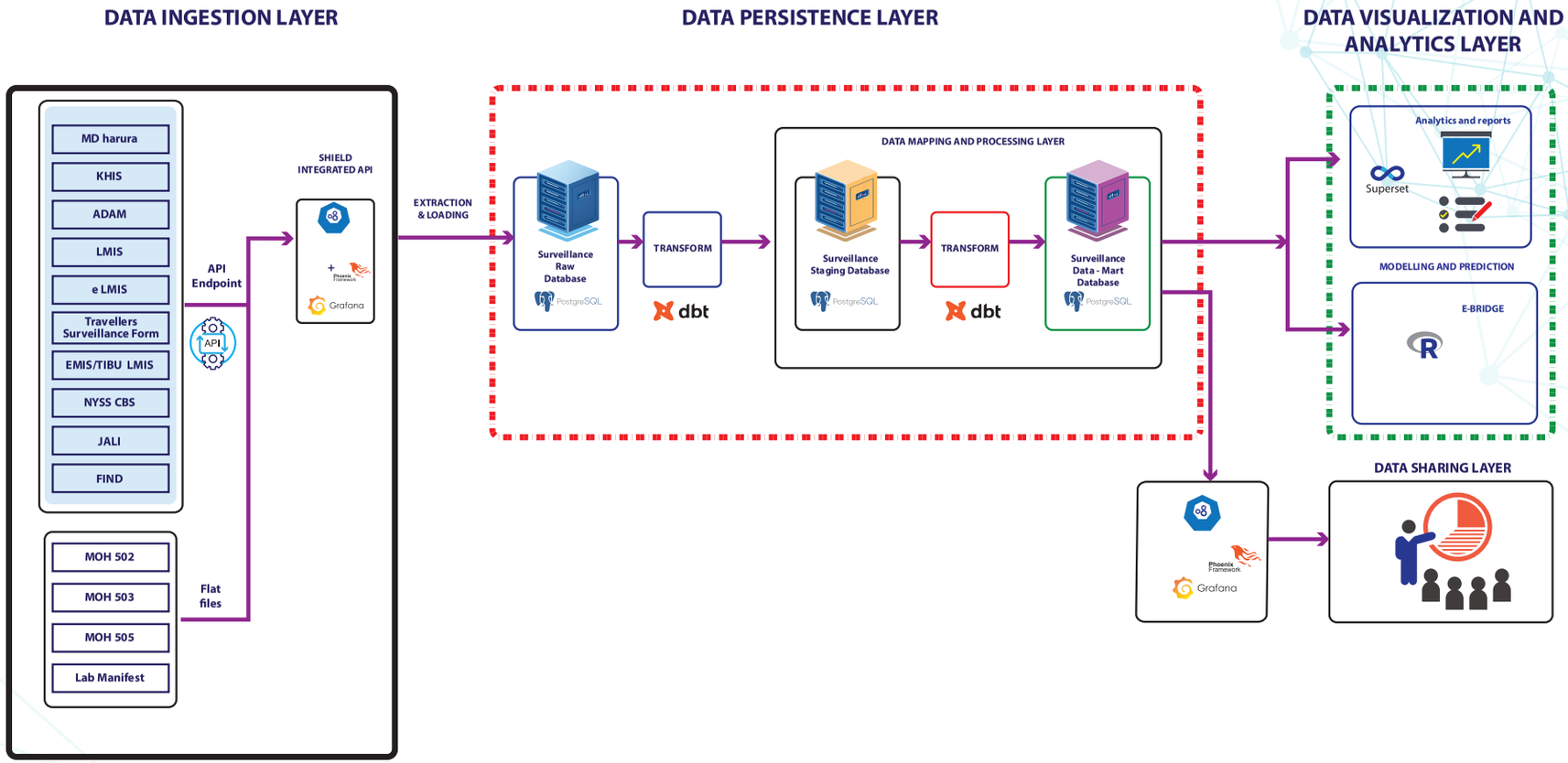


Figure 8: National Disease Surveillance Repository Architecture

The National Disease Surveillance Repository will be a key component of the e-surveillance ecosystem since it will host diverse datasets from source surveillance systems and avail it for use in disease surveillance and outbreak response efforts. Implementation of activities under all thematic areas including data integration and digitisation efforts are tied to its evolution, as such development of the National Disease Surveillance Repository will be coordinated with that of source systems ensuring that they are aligned to avoid delays in availing important surveillance datasets from digital solutions for use in public health surveillance.

National Disease Surveillance Repository Roadmap

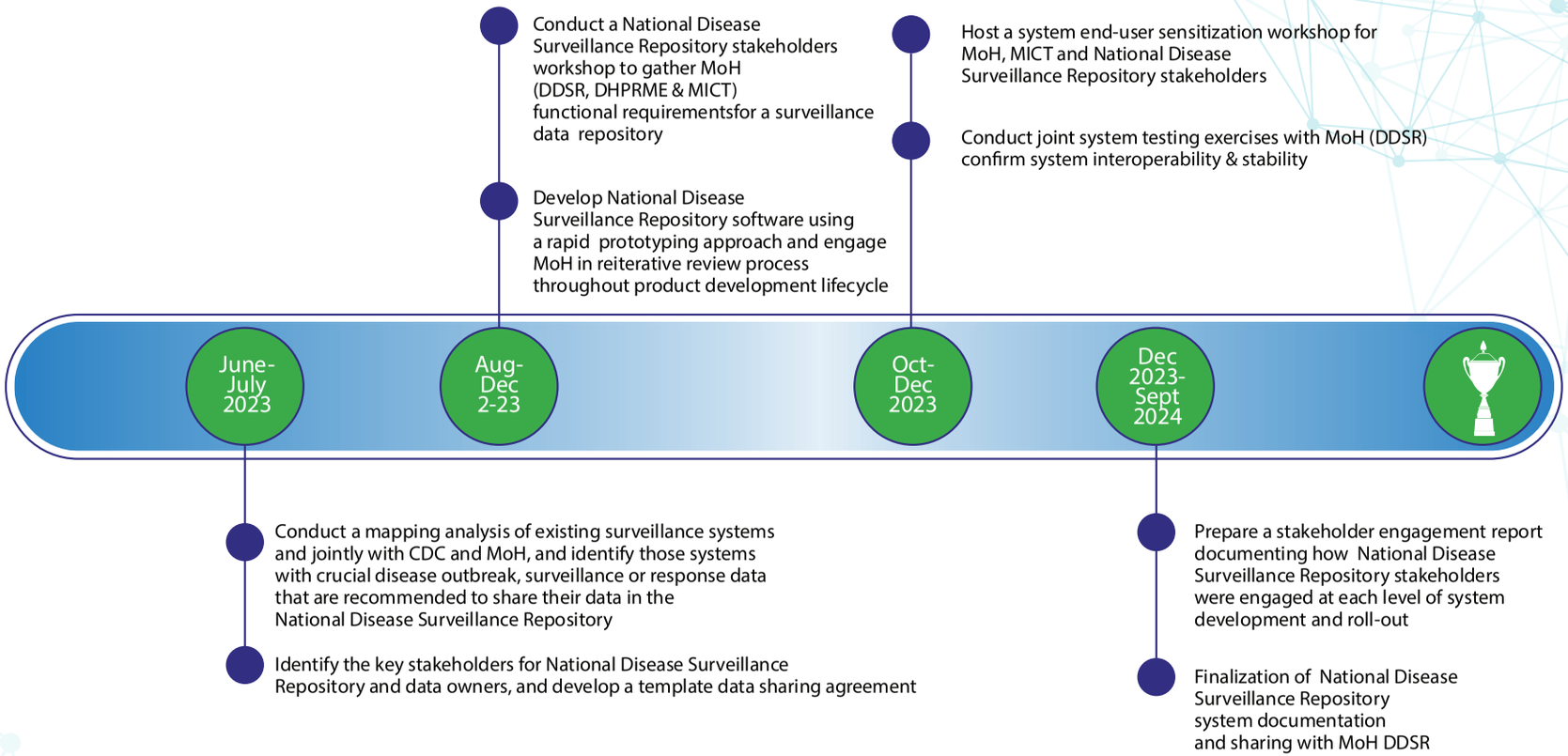


Figure 9: Proposed Implementation Roadmap for the National Disease Surveillance Repository

3.2.2. Thematic Area 2: Revitalise Routine Surveillance by Digitising Transmission and Hosting of Disease Surveillance Data in a Central Surveillance Repository.

Routine Surveillance Systems provide background data on disease patterns that are critical to intelligence for outbreak detection and response. The overall strategy adopted by KNPHI is to strengthen routine surveillance by digitising data management for surveillance efforts that are currently conducted manually, streamline data collection and transmission for already digitised disease surveillance efforts, retire duplicate non-standard digital solutions used for disease surveillance. Additionally, KNPHI will progressively map and ensure all disease surveillance datasets routinely collected through digital solutions are documented and transmitted to a centralised surveillance repository.

3.2.2.1 Focus Area 2.1: Transmission of Routine Disease Surveillance Data to the National Disease Surveillance Repository

The Ministry of Health collects disease surveillance data using several digital health solutions and tools. Most of this data is collected and stored in distributed databases that are not interconnected, the servers where the data is located are often not within MOH infrastructure or within MOH control. To resolve this, the KNPHI and its stakeholders have developed an architecture, which will ensure that data from the various source systems will flow the National Disease Surveillance Repository as outlined in the figure 12 below. KNPHI will also maintain an inventory of all surveillance data collection systems and progressively ensure that they transmit routinely collected data to the national repository. This will ensure that surveillance datasets are in the custody of the MoH and accessible at all levels for decision-making.

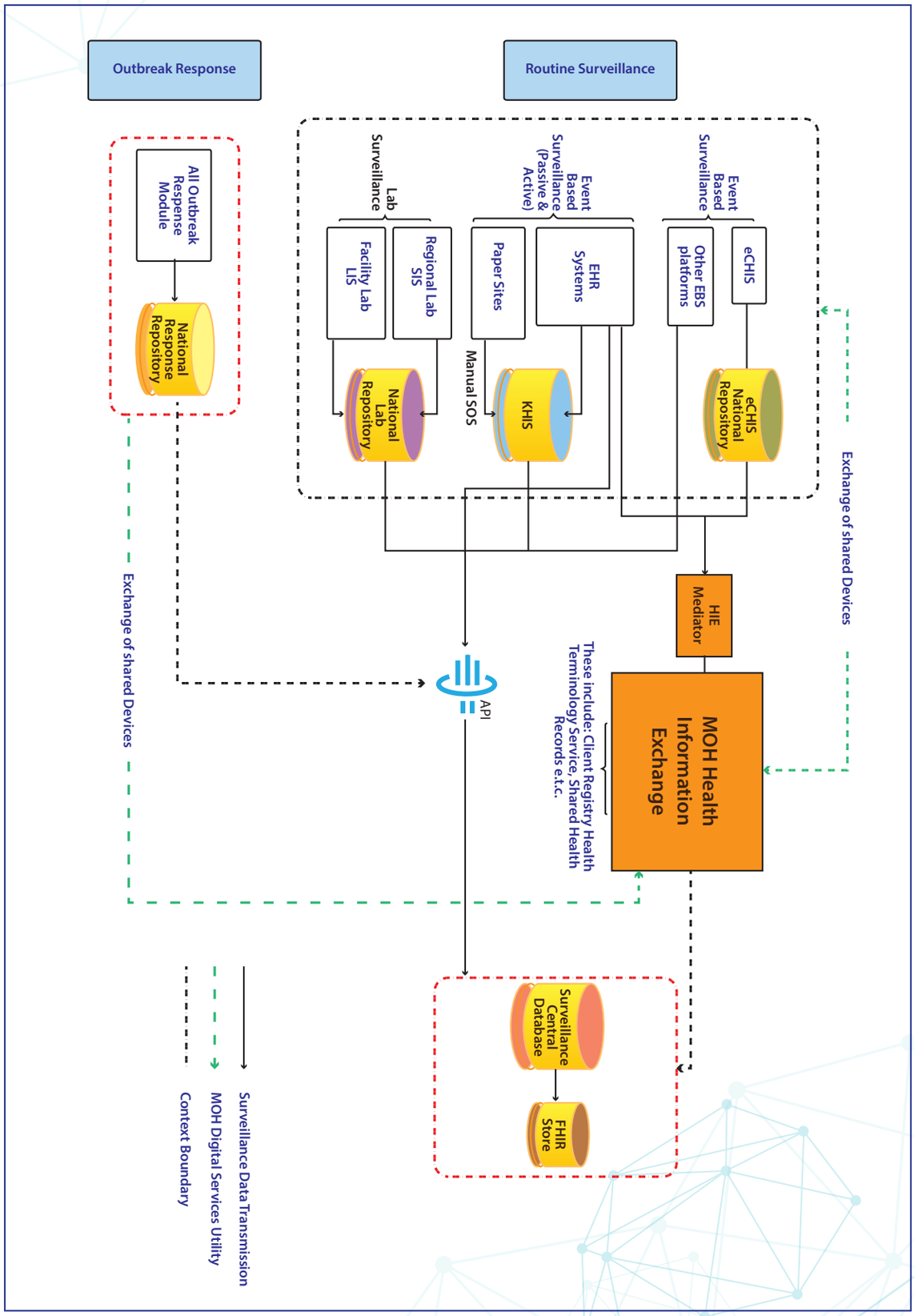


Figure 10: Flow of Surveillance Data from Source systems to the National Disease Surveillance Repository

The e-surveillance architecture workflow can be described as follows:

1. Disease Surveillance applications are classified according to the type of surveillance they support. This classification is system-agnostic and accommodates all systems operating within the specific surveillance area.

2. Disease Surveillance data is collected and organised by the source digital systems and then shared with the national repository. Most of the applications are capable of FHIR-standard-based interoperability with other digital systems. They are expected to transmit surveillance data through mediators such as OpenHIM and through which they will also access services from MOH-shared utilities such as Client Registry (CR), National Health Terminology Services (NHTS) and the Shareable Health Record (SHR).

3. Selected source applications will also transmit data to National Disease Surveillance Repository via the sync API. The applications will also leverage MOH digital health services such as Unique Person Identifiers (UPI) from the Client Registry allowing for unique identification of patient records.

4. The disease surveillance data submitted, and National Disease Surveillance Repository will be staged stored, processed, and analysed. The outputs will take the form of relevant dashboards and reports built in integrated business intelligence tools as well as alerts/messages that can be transmitted back to the surveillance applications, e.g., outbreak response applications.

System owners for this phase of the application will be required to provide additional information and documentation about the existing application to facilitate planning for transmission. Information required includes:

- Technology used, security and hosting information.
- Datasets collected,
- Data dictionary mapped to national health terminology services metadata,
- Standards for data exchange supported,
- API documentation where applicable

For this strategy, KNPHI and stakeholders have mapped all digital systems currently used to collect surveillance data. The KNPHI has adopted a phased approach to data transmission that will ensure that all surveillance data is identified and progressively transmitted to National Disease Surveillance Repository in real time. Several solutions were prioritised for the first phase.

Phase I of data transmission and integration will involve ensuring routinely collected data from the following routine surveillance and laboratory surveillance applications listed below are transmitted to National Disease Surveillance Repository.

It is expected that at the end of this objective, the KNPHI will have consolidated all relevant historical surveillance datasets into a single repository and initiate routine submission of disease surveillance data into the National Disease Surveillance repository. Upon transmission of historical data. KNPHI will review existing applications that generate different disease surveillance datasets. The applications will be assessed to determine the level of compliance with the digital health certification framework and where possible, supported to comply. A triaging process to identify duplicative data collection applications, redundant data collection processes will be documented, and a transition process initiated. A comprehensive plan and process of replacing data collection solutions that are unable to comply with certification standards, digital health/data legislation and compliance with MOH approved hosting plans that include data sovereignty requirements will be conducted.

3.2.2.2.Focus Area 2.2. Digitisation of Previously Manual Routine Surveillance Initiatives

Data collection for most surveillance initiatives is currently done through manual, paper-based systems. These manual data management systems, that also affect components of IDSR, contribute to most of the inefficiencies in surveillance data collection and management. The KNPHI and its partners will identify, map, and progressively adopt digital solutions for digitisation of these manual data collection efforts. For this, KNPHI will leverage the digital transformation currently being implemented in the sector to strengthen events-based surveillance data collection using through ECHIS, indicator-based surveillance

data collection and reporting through EMR/EHR generated reports and additional surveillance data collection tools such as LIS for lab surveillance. All data emanating from newly digitised surveillance efforts will be transmitted to National Disease Surveillance Repository as outlined in focus area 2.1. The manual tools prioritised for this focus area are displayed in Appendix 3.

3.2.2.3. Focus Area 2.3: Leveraging Digital Solutions in the Implementation of New Surveillance Approaches (Passive and Syndromic Surveillance; Environmental Surveillance; Food Surveillance; and One Health Approach)

To overcome the challenges presented by the manual process currently employed for syndromic (ILI/SARI) and passive surveillance, the KNPHI will leverage point of care clinical care EMRs/EHRs for collection of passive surveillance and syndromic surveillance data. Building on the successful prototype developed for SARI and ILI and implemented at sentinel surveillance sites, the KNPHI will leverage the digital transformation in health to recruit new health facilities to participate in passive surveillance. The division will and adopt passive surveillance for new priority disease areas.

By incorporating passive surveillance workflows alerts case detection and reporting features within the EMR, disease surveillance data can be collected seamlessly during routine clinical care provision to identify and flag potential cases for diseases under surveillance. For the first phase, KNPHI will leverage county procured EMR systems used in SARI and ILI facilities for passive surveillance and expand to facilities running similar systems.

Environmental surveillance and disease surveillance are intricately linked in their efforts to protect public health. Environmental surveillance involves monitoring and analysing the physical, chemical, biological factors and in our surroundings. It plays a pivotal role in disease prevention and control by detecting environmental factors that can influence the spread of diseases. Food surveillance is a fundamental component of public health and regulatory efforts to prevent foodborne illnesses and promote safe and nutritious food choices.

The KNPHI will pull processed signals from Ministry of environment and food surveillance into National Disease Surveillance Repository and overlay for detect unusual patterns or events, serving as an early warning system for disease outbreaks. By integrating information from both environmental and disease surveillance, public health authorities can identify and mitigate health risks more effectively. A table displaying the digital tools prioritised for the first phase of transmission is displayed in Appendix 2.

3.2.2.4 Focus Area 2.4: Strengthening and Digitisation of Information Sharing Between Animal Health and Human Health Sectors.

To enable timely and information sharing mechanism among human health and animal health players, new data sharing APIs should be established within the National Disease Surveillance Repository that draws data from human, environmental and animal health sectors. An analytics module will be established jointly with the Environmental Health teams from Ministry of Health and Ministry of Environment, Animal Disease surveillance teams from Ministry of Livestock and Ministry of Wildlife under the Directorate's Zoonotic Disease Unit to facilitate sharing of indicator-based data and case level data from the three domains. Meaningful linkage of animal and human surveillance efforts through the One health Approach will be achieved through:

- Progressive automation through adoption of easy-to-use interoperable ICT solutions to digitise manual reporting disease surveillance initiatives, especially in livestock and wildlife.
- Adoption of technology-enabled data sharing on human, livestock, wildlife, environmental surveillance to support more efficient surveillance and response to zoonoses and other public health threats under one health initiative as described above.
- Establishment of a One Health portal that consolidates human and animal disease datasets, data analytic outputs, reports, and publications.
- Strengthening of specimen referral structures through digitisation.

There exists a strong coordination structure within the ZDU that provides a central point for broad based coordination of different experts from the three Ministries to implement the One Health Strategic Plan.

A comprehensive integrated Early warning system will be built within the data sharing portal that displays clear thresholds and automated alert system whenever the occurrence of disease or specific events are observed in any of the three domains. This early warning system trigger visual, messaging alerts at all the nodes and levels of surveillance from sub county levels to the `National level for all the sectors.

Both the dashboards and reports generated from data in the developed platform should show trends of the six priority diseases and updates of current outbreaks of the six priority diseases. Stakeholders will be facilitated to generate datasets from the platform and share the reports/ bulletins with relevant stakeholders in the animal health, environmental health, and human health sectors.

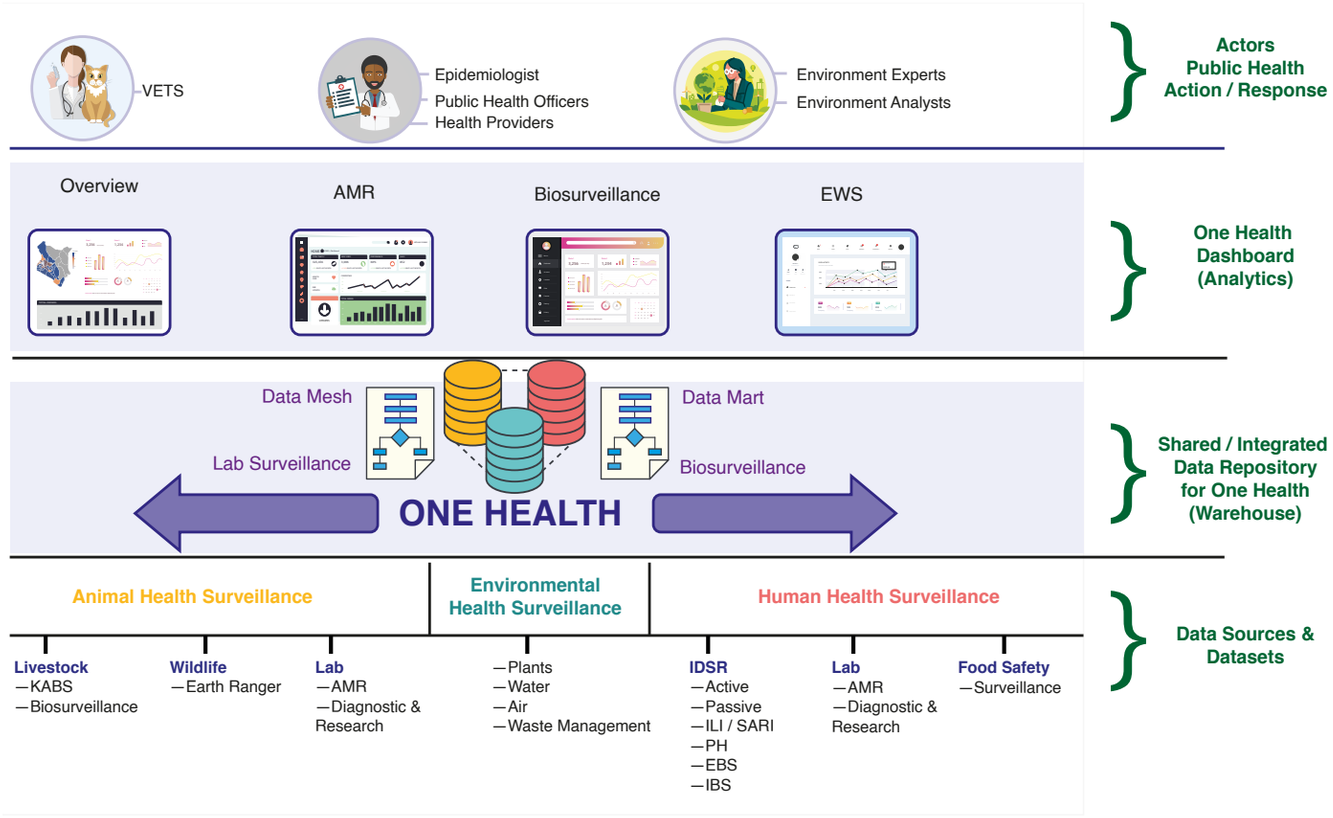


Figure 11: Envisioned One Health Functional Architecture

3.2.2.5.Focus Area 2.5: Adoption of Unique Patient Identification for Routine and Outbreak Response Surveillance Efforts.

Outbreak Response involves generating line-lists individuals infected or affected by public health events. Additionally, routine surveillance systems that capture case level data such as laboratory Information Systems, Electronic Medical Record Systems must be able to uniquely identify individuals affected by events of public health importance for the purpose of linking epi datasets with lab datasets, contact tracing and to reduce duplicates in reported cases. In case of an outbreak the cases reported ought to be unique to provide an accurate picture of the situation. Additionally, it is necessary to track a client's movement and identify their contacts. National Unique Patient Identifier presents a mechanism to identify cases and contacts. The MOH Client Registry has been developed as a central source of truth on the identity of clients. Together with a Health Worker Registry. The role of the MOH client registry will be to facilitate authentication of persons interacting with the health system (either as a patient or a care provider) across the country.

The KNPHI will develop surveillance business case for Unique Person Identification, map the surveillance systems that collect case level data, assess them, and facilitate access to the MOH Client Registry for Unique Person Identification. Selected applications will be enhanced to enable recording of Nationally issued Identifiers, pull MOH issued unique person identification number and ensure these can be used for deduplication, contacts management and for linking datasets used for lab-epi interface. Systems that generate and pull shared Health Record will be utilised across different systems to facilitate case identification, investigation, and reporting. Some use-cases for NUPI for Disease Surveillance are outlined below.

- Deduplication of case reports; by alerting surveillance officers of persons reported to the central repository from different facilities,
- Historical records retrieval and linking with new cases to identify re-infections,
- Support in contact tracing and management where applicable,
- Linking datasets from different clinical settings/ facilities,
- Strengthening Lab-epi interface during routine surveillance
- Linking Case Investigation datasets with lab orders and results during outbreaks

3.2.3 Thematic Area 3: Strengthen Lab-Epi interface Through Automation of Lab Sample and Results Management.

An integral component of surveillance involves seamless electronic specimen labelling and logging of laboratory orders from suspected cases to laboratories during routine surveillance and outbreak response. The automation of lab order-to-results workflow ensures accuracy in linking epidemiological data from cases under investigation with their laboratory results. Implementation of a structured interaction between participating routine surveillance, outbreak investigation and laboratory systems will ensure a smooth flow of information. The diagram below details the proposed flow of information between the participating clinical and laboratory systems. To realise this goal, KNPHI will define a standard minimum dataset for lab orders, design data transmission channels for remote logging of lab requests for routine surveillance/outbreak investigation and for returning results to surveillance and outbreak response applications.

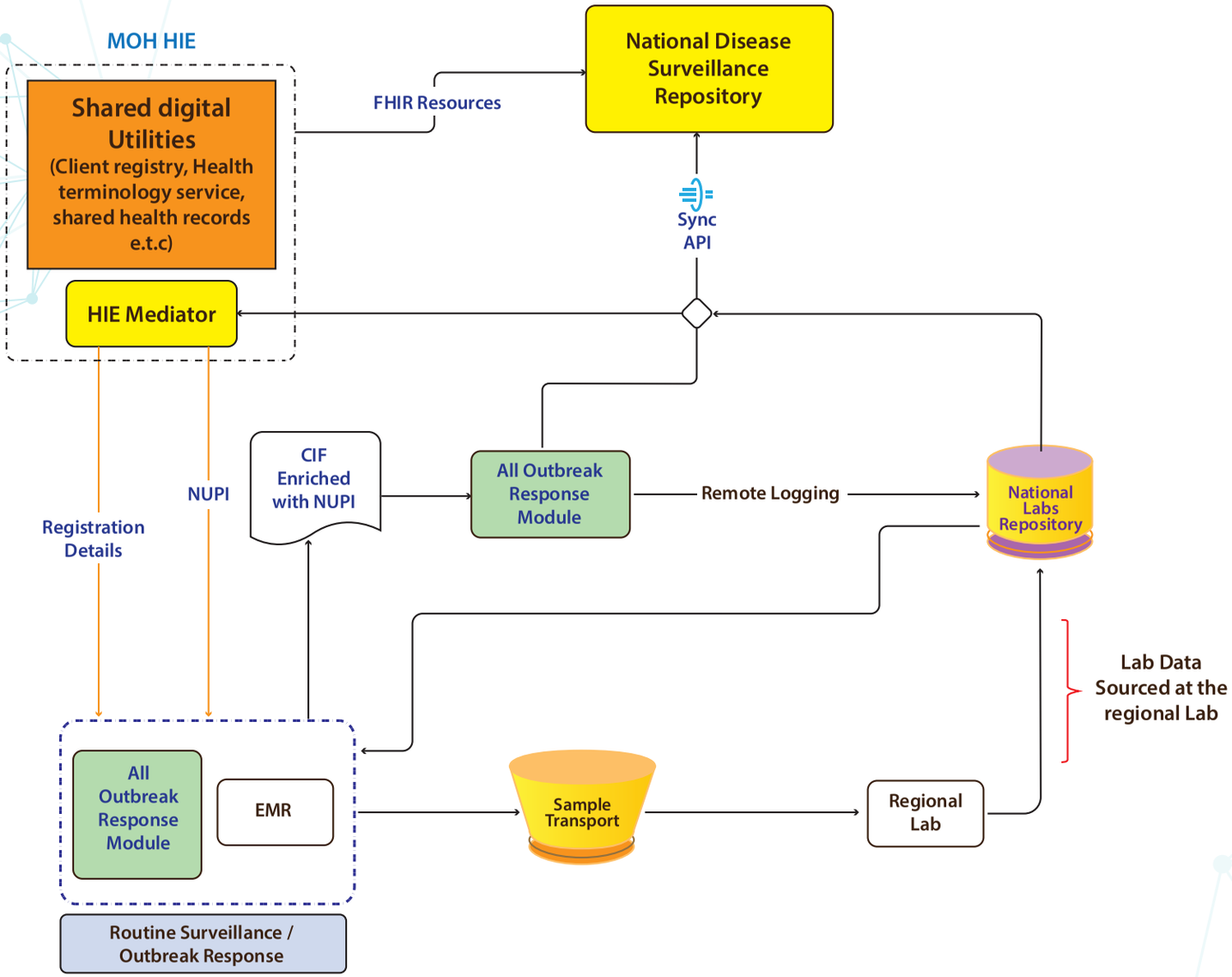


Figure 12: Envisioned One Health Functional Architecture

Laboratory-Epi Interface Architecture components:

- MOH National Digital Health Services: A suite of platforms providing digital health services under the Kenya National Health Information Exchange e.g., client registry, health terminology service.
- Outbreak Response solution: An end-to-end platform for tracking new and re-emerging disease outbreaks.
- EMR: Electronic Medical Record systems that handle collection, storage, and processing of patient health records.
- Regional Laboratories: Specialized standalone laboratories that handle samples above the facility level e.g., AMR samples.
- National Laboratory Repository: A central database that holds consolidated lab data from disparate lab management systems.
- Sync API: A standardised Application Programming Interface that serves as the entry point of surveillance data sent to the National Disease Surveillance Repository.
- National Disease Surveillance Repository: A centralised repository to support data storage and analysis of all surveillance data.

The architecture workflow can be described as follows:

1. Surveillance applications generate a Case Investigation Form (CIF) with the patient registration, clinical and epidemiological details.
2. Case Investigation forms are submitted to the MOH client registry which returns the patient's NUPI. The issued NUPI together with the unique case ID offer unique identifiers for both the patient and investigated case respectively.
3. The CIF collected by user facing end of the selected outbreak response solution is then uploaded to a central database where it is staged and further transmitted to National Disease Surveillance Repository. At this stage, the lab sample labelled with the case ID and NUPI defined in the CIF is also transported physically to the regional lab for analysis.
4. The regional lab then submits the relevant case information to the national lab repository. An integration between the national laboratory repository, the outbreak response solution and the National Disease Surveillance Repository allows for routine pulling of lab data into National Disease Surveillance Repository which can link datasets from all three sources to the same case.
5. The outbreak response solution is also able to pull data from the national labs repository and picks up lab results relevant to the various uploaded CIFs. These results can be passed downstream to the surveillance applications as well as upstream to National Disease Surveillance Repository via the Sync API.

3.2.4 Thematic Area 4: Strengthen Outbreak Response Capability by Digitising Outbreak Response Workflows.

Rapid and effective response to a public health emergency such as a suspected outbreak or other public health event not only calls for an immediate response but is also one of the core capacities required by the Public Health Emergency Management Committee (PHEMC) in support of the Universal Health Coverage, Vision 2030, and International Health Regulations (IHR 2005). Being prepared to detect and respond to such an event is an essential role of the Community, health facility, Sub-County, County and National levels.

3.2.4.1. Focus Area 4.1: Develop an Outbreak Response Module

Several systems have been deployed during past outbreaks to support response efforts. This has proven inefficient and unsustainable, since most of these applications serve during the outbreak period and are then retired, never to be used for similar outbreaks. Additionally, datasets from these outbreaks end up in different locations and therefore become inaccessible.

To address this gap, the Kenya National Public Health Institute will develop an All-Disease Outbreak Module. The developed solution will be an end–to-end outbreak response platform for investigating, line-listing, and tracking new and re-emerging disease outbreaks, The solution will have in-built standardised and disease specific case investigation workflows and tools making it reusable with little effort whenever multiple outbreaks for one disease occur. It will host generic case investigation tools/forms that can be deployed quickly in response of new and re-emerging disease outbreaks. This will ensure that outbreak responses are fast, digitally conducted, and the data collected is centralised for a coordinated response.

Upon completion of the outbreak response solution, KNPHI will digitise Case Investigation Forms (CIF). The first version of these forms will be based on WHO approved Case Investigation tools. The KNPHI has prioritised diseases that have had the most recent outbreaks digitisation in Phase I.

In prioritising which forms to be digitised KNPHI adopted a syndromic approach will be guided by IDSR diseases, the Ministry of health adopted a syndromic approach. This approach involved initially grouping disease conditions into disease syndromes to ensure few forms can be used to investigate multiple diseases early before CIF for all disease conditions are developed. Examples of the syndromes considered include:include Influenza like illnesses (ILI) Severe Acute Respiratory Illnesses (SARI), gastro-intestinal Illness (GII), neurological illness, fever & rash illness, and viral haemorrhagic fever, among others. Other criteria considered include:

1. Potential Public Health Impact: The KNPHI considers the potential impact of the disease on public health interest in case of an outbreak. Diseases with high morbidity, mortality, or epidemic potential may require more urgent and automated surveillance efforts.
2. Disease Characteristics: The nature of the disease, including its transmission dynamics, incubation period, and detectability, can influence the feasibility and design of automated surveillance systems. In this regard, diseases with highest potential for outbreaks, especially those that featured frequently in recent outbreaks will be prioritized.
3. Diseases earmarked for elimination/ eradication: Diseases such as Acute Flaccid Paralysis/polio that are earmarked for eradication/elimination will be prioritised.

The first phase involves digitisation of CIFs for covered about 15 disease conditions selected using this criterion. Upon completion of this phase, the KNPHI and its partners will progressively digitise CIFs, with the goal of introducing custom CIFs for all the IDSR priority diseases in preparation for the next outbreak.

Tracking event data is important because some event characteristics (e.g., reporting or response dates) are not captured in case-level data. Previously, there has been no systematic way to track outbreaks in Kenya, which has hampered opportunities for performance improvement through approaches like 7-1-7. The adopted outbreak response solution will help address this gap.

3.2.5. Thematic Area 5: Strengthen Data Governance Structures and Facilitate Data Sharing and Use for Public Health Surveillance.

Surveillance involves generation and use of individual-level data for public health response. The passage of the National Data Protection Act, 2019 (DPA) introduced legal safeguards for protection of Personally Identifiable Information (PII) data. This necessitated the development health specific guidelines that promote use of data for routine public health action with the need for data protection. It is therefore imperative that the relevant data governance structures are put in place to guide the sharing and use of surveillance data in a proper and ethical manner and discourage hoarding of valuable disease surveillance datasets.

The Ministry of Health has coordinated stakeholders through a participatory process of developing the Health Data Governance Framework (HDGF). The framework provides guidelines of ethical and secure sharing of sensitive personal data and de-identified health data including disease surveillance data. While protecting data subjects is a critical part of KHDGF, as outlined in the Kenya's DPA while considering unique health care settings, the frameworks also require MOH to promote sharing of data for public health interventions.

A fundamental component of the HDGF is the Kenya Health Enterprise Architecture (KHEA) blueprint. The KHEA blueprint defines how different systems will be organised and will interoperate to realise health and data objectives. It is a foundational document to the

HDGF and defines how source systems will interact with reporting solutions, such as the National Disease Surveillance Repository and the analytics platform, to deliver value of public health datasets to public health and clinical care teams at all levels. The MOH in collaboration with surveillance and other stakeholders published a Digital Health Act in 2023 as well as regulations that provide a legal basis for the health data governance framework. Awareness creation and capacity building will be conducted among the actors in the disease surveillance context as part of efforts towards operationalisation of these legislative and policy documents to ensure early adoption. KNPHI and its partners will collaborate with the Digital Health Agency in reorganising the surveillance ecosystem to align with the new law and regulations.

Key Activity	Sub-Activity	Short Term (2024-2025)	Medium Term (2026-2027)	Long Term (2028-2029)
Launch of Document	• Develop a comprehensive and user-friendly (aka popular version) document outlining the health data governance Framework (popular version).	X		
	• MoH to lead officail launch of Framework.	X		
Dissemination and Sensitisation of State and non-State actors	• Develop a communication and dissemination plan to reach out to all State and non-State actors. • Conduct engagement sessions with State and non-State actors to sensitize them on the benefits and the importance of the health data governance Framework, and how it will improve the quality of healthcare.	X		
Operationalisation of Data Governance Structures	• Operationalise the governance structures as out-lined in the governance section of this policy document in alignment with the partnership Framework. • Establish data governance structures at the national, county, and facility levels, as stipulated in the governance section of the health data governance Framework document. • Develop job descriptions for data governance roles at each level and recruit qualified personnel to fill those roles.	X	X	X
Health Data Governance Sharing and Learning Forum	• Organise a health data sharing and learning forum to share best practices, experiences, and challenges related to the implementation of the health data governance Framework. • Engage stakeholders in interactive sessions to develop actionable plans to address emerging issues in health data governance.	X	X	X
Development and operationalisation of Health Data Governance Tools	• Establish sector registry of health data controllers and processors. • Develop standard Procedures to operationalise data governance Framework. • Operationalise standard templates for data governance. • Develop monitoring tools for process monitoring and evaluation.	X	X	X

Figure 13 Health Data Governance Framework Roadmap

3.2.6. Thematic Area 6: Deployment of the Necessary Infrastructure (hardware and software) Needs for Operationalising a National Disease Surveillance Repository.

The national Ministry of Health has developed a national data center to consolidate data generated by health facilities in a secure MOH-managed repository. The establishment of the data center is one of the steps towards aligning health data management with the Kenya Data Protection Act of 2019. The DMI funding will support in providing technical assistance to the MOH towards the development of the governance structures and policies for the data center and providing critical infrastructure pieces required for optimal operations of the data center. This will play a critical role in ensuring the surveillance repository is hosted within the MOH data center. An optimally functioning data center will enable data driven initiatives benefit from this resource with minimal disruption of clinical and surveillance activities. The MOH will provision servers, with processing, storage, and secure internet connection to enable hosting of surveillance applications in the MOH environment and facilitate secure transmission of surveillance datasets to the central surveillance repository. Additionally, the KNPHI will establish a cloud back-up of the surveillance repository to maintain optimal continuity of service and avoid down-times.

3.2.7. Thematic Area 7: Develop a Workforce for Effective Routine e-Surveillance and Outbreak Response.

3.2.7.1. Focus Area 7.1: Strengthening the Capacity of Rapid Response Teams (RRTs) Stationed in National and County Emergency Operations Centers (EOC).

Outbreaks are often unexpected and can be unpredictable. Lack of easy-to-use digital tools and inadequate capacity to adopt, customise and deploy an outbreak response digital system during outbreaks remains a challenge. To overcome this challenge, KNPHI will train Rapid Response Teams (RRTs) on the newly developed outbreak response tools in preparation for future outbreaks.

Upon completion of outbreak response applications and digitisation of Case Investigation Forms envisioned in Phase I, KNPHI will conduct capacity strengthening of RRTs on approved outbreak response solution equipping them with the knowledge they need to quickly adopt and activate outbreak response solution for ongoing and new outbreaks. To achieve this, the KNPHI will:

- Update the FELTP frontline epidemiology course with training content on All Outbreak Response Module, National Disease Surveillance Repository, new public health surveillance dashboards, and relevant informatics training.
- Develop an inventory of active and reserve RRTs stationed Nationally and in Counties
- Conduct a Training Needs Assessment (TNA) among Active RRTs stationed within the Emergency Operations Center
- Guided by the TNA, conduct Frontline and intermediate FELTP trainings targeting RRTs and PHEOCs to cover gaps in uptake of the Frontline FELTP course and the new surveillance and outbreak response tools including the outbreak response solution and National Disease Surveillance Repository.
- Ongoing training of staff in the Public Health Emergency Operations Centers on the new data flows, tools in use for routine public health surveillance, laboratory-based surveillance, and outbreak response tools. The capacity of PHEOC staff should be developed and strengthened to empower them to use newly created dashboards, equipment and software deployed within the PHEOCs.
- The training platforms such as the MOH virtual academy, universities' e-learning forums, other in-person and virtual training platforms should be adopted.

3.2.8. Thematic Area 8: Digitise Public Health Emergency Operations Centers (PHEOCs) for Effective Detection of Public Health Events, Preparedness and Response.

The Kenya PHEOC was inaugurated on 17th May 2016 by the Cabinet Secretary for Health and the World Health Organisation Country Representative. The PHEOC's mandate is to coordinate the implementation of this emergency management plan. The national PHEOC is part of a network of national and international Emergency Operating Centers (EOCs). Several PHEOCs have been established at county level in close collaboration with Ministry of Health, these PHEOCs will work in concert with the national

PHEOC during both preparedness and response phases guided by the Concept of Operations (CONOPS).

Emergency operations centers operate within the WHO framework for PHEOCs. WHO framework for the incident Management System, consisting of Management, planning, operations and logistics, finance and admin will guide its activities. PHEOCs handle events-specific data, events management data and context data. PHEO must plug into new and existing data sources including the National Disease Surveillance Repository and the Digital Superhighway to access relevant datasets to guide epidemic response interventions.

ICT infrastructure is one of key requirements during setup and operationalization of PHEO. Some key infrastructural considerations include a communication system that ensures robust internal and external communication (incl radio and satellite telephony), screens/displays for visual representation of events, Media monitoring capabilities. Operational computing equipment terminals, printers, and networks.

Principles used in this section are based on the WHO framework for PHEOC that promotes Country ownership, building on existing initiatives, implemented incrementally, human resource and capacity strengthening.

3.2.8.1. Focus area 8.1: Adopt Incident Management Software to Strengthen Response Capabilities of National and Regional Public Health Emergency Operations Centers (PHEOCs)

Establish a public health operation center to streamline Public Health Emergency Operations; specifically, the Joint Information Center (JIC) which is a central location where public health information is coordinated, processed, and disseminated to the public and media. It works closely with the PIO to ensure consistent messaging. These public health emergency operations centers can be physical locations or a virtual center.

Integrate a data management system to collect, analyse, and share data relevant to incident management. The system will draw, process, and display structured and unstructured data from diverse sources including routine surveillance data from the National Disease Surveillance Repository, Laboratory repository and outbreak management information, social and mainstream media to guide response interventions. The dashboard will enable real-time data visualization, enhance situation awareness, guide resource allocation and planning, enhance collaboration and risk communication. Establishment of the Information Systems, communication, back-up, and cybersecurity capabilities of PHEOCs should be guided by the Kenya Public Health Emergency Operations Center Handbook.

3.2.8.2. Focus area 8.2: Adopt Digital Communication Tools to Strengthen Collaboration and Communication in PHEOCs.

A robust communication system will be established in the public health emergency operation center. This will be a central communication structure that is accessed by County Public Health Emergency Operations Centers. A centrally managed, federated service desk (tools, people, and process) will be established to facilitate communication between national, county, sub-county, facility, and community levels. The communication system will adopt and leverage other communication tools such as messaging apps, video conferencing, and collaboration platforms. A notification system capable of operating optimally during the watch-warn-alert modes will be adopted. It will have an early warning alert system that triggers responses to new and active public health events with SOPs for activating response interventions.

3.2.8.3. Focus area 8.3: Enable Establishment and Activation of an Emergency Response Supply Chain Management Strategic Stockpiles' Rapid Response to Public Health Emergencies.

The National Public Health Emergency Operations Center will adopt and implement a commodity management software linked to accredited suppliers through the envisioned National Logistics Management Information System. This will ensure rapid mobilisation of strategic stockpiles of commodities in response to public health events when needed at both national and county levels.

4

Roadmap to Digital Public Health Surveillance



To attain the desired future state where e-Surveillance is functional and used by stakeholders, a phased development and deployment of the solution will be adopted. Below is a summary of the milestones, activities, responsibilities, and timeframes for Phase I implementation.

A detailed workplan for this phase is provided separately. Each year, KNPHI will develop an annual workplan outlining activities towards the delivery of the remaining phases of this strategy.

Milestones and Activities	Responsible Organisation	Time Frame				
		2024	2025	2026	2027	2028
Thematic Area 1: Development of a National Disease Surveillance Repository at the Ministry of Health to support data driven decision making.						
Focus Area 1.1: Development of National Disease Surveillance Repository and visualizations						
Development of the National Disease Surveillance Repository database.	**MOH, IPs**	X	X			
Development of APIs to receive data into the National Disease Surveillance Repository from participating systems.		X	X			
Enhance the National Disease Surveillance Repository for FHIR capabilities to support integration with other platforms in the MOH eco-system.		X	X			
Identify appropriate platform to support visualization of surveillance data from repository.		X				
Training and sensitization on the National Disease Surveillance Repository and visualisation tools.			X	X	X	X
Thematic Area 2: Strengthen Routine Surveillance through digitisation, transmission, and hosting of disease surveillance data in a central surveillance repository						
Focus Area 2.1: Transmission of routine disease surveillance data to the National Disease Surveillance Repository						
Map all systems currently used for routine surveillance and Lab surveillance, hosting, ownership, and surveillance efforts they support.	**MOH IPs System Owners**	X				
Update technical documentation of all routine surveillance solutions including data dictionaries, metadata, and interoperability capabilities.		X	X	X		
Enhance the identified routine surveillance and lab solutions to transmit data to the National data repository using a phased approach.		X	X	X	X	X
Focus Area 2.2: Digitisation of previously manual Routine Surveillance initiatives						
Identify the targeted manual surveillance tools for digitisation in line with the established structures.	**MOH IPs System Owners**	X				
Enhance existing systems to digitise targeted manual surveillance tools.			X	X	X	
Develop ETL structures within the existing systems extract and transform surveillance data within the platform.		X	X			
Enhance existing systems to integrate into API interfaces from the national surveillance to transmit surveillance data.			X	X	X	X

Focus Area 2.3: Leveraging digital solutions in the implementation of new surveillance approaches (Passive and Syndromic Surveillance; environmental surveillance, Food surveillance and One Health approach)						
Conduct sensitization meeting with MOH approved solutions and county procured system owners on potential incorporation of passive disease surveillance workflows in EMR.	**MOH IPs**	X	X			
Enhance MOH approved solutions and county procured system owners to implement enhancements identified to incorporate passive disease surveillance.		X	X			
Training and sensitization on enhanced systems incorporated with passive disease surveillance.			X			
Focus Area 2.4: Adoption of Unique Patient Identification for routine surveillance and outbreak response efforts						
Develop and document business case and value proposition for use of NUPI for disease surveillance and identify priority applications that will access Client Registry.	**MOH IPs System Owners**	X	X			
Conduct engagements with MOH on wider extension of NUPI to support disease surveillance.		X	X			
Convene Meetings with stakeholders for awareness creation of NUPI and potential use-cases for NUPI integration to support disease surveillance.		X	X			
Identify priority applications and enhance them to implement NUPI use-cases for implementation.			X	X	X	
Thematic Area 3: Strengthen Lab-Clinical Interface through automation of Lab sample and results management						
Conduct requirements gathering and analysis for data exchange use cases for the different clinical care systems and laboratory systems.	**MOH IPs**	X				
Support the design of integration points and standards for data exchange of clinical and surveillance datasets amongst clinical and lab systems.		X	X			
Support the adoption and development of the interoperability solution between the different clinical and lab systems.			X	X		

Table 3: Proposed Phase 1 Digitization Workplan

Thematic Area 4: Strengthen Outbreak Response capability by digitising outbreak response workflows						
Mapped out diseases of interest which are to be prioritised for digitisation in the response tool.	**MOH IPs**	X				
Enhance outbreak response tool with case identification forms for diseases of interest.		X	X			
Transmit data from the outbreak response tool to the National Disease Surveillance Repository.			X			
Conduct training and capacity building of the outbreak response tool to stakeholders for use.			X			
Thematic Area 5: Strengthen Data Governance Structures and facilitate data sharing and use for public health surveillance						
Support MOH in the development of the HDGF.	**MOH IPs**	X	X			
Develop SOPs to guide data access and dissemination of the HDG framework.		X	X			
Conduct engagements to sensitize stakeholders on the HDG framework.		X	X			
Support MoH in developing data sharing agreement templates for ease of data exchange.		X	X			
Participate and provide feedback on eHealth (Digital Health) Bill in readiness of setting up systems to support disease surveillance.		X	X			
Thematic Area 6: Support the deployment of the necessary infrastructure (hardware and software) needs for operationalizing a National Disease Surveillance Repository						
Review resource requirements for cloud readiness at the NDC.	**MOH IPs**	X	X			
Deploy and setup infrastructure at the NDC to support cloud readiness.			X			
Operationalization of the SOPs for the NDC.			X			
Capacity build and provide Technical Support in the operation & management of the national data center.			X			
Support MOH to develop guidelines and policies for cloud backup at the NDC.			X			

Thematic Area 7: Workforce development for effective routine e-surveillance and outbreak response						
Focus Area 7.1 Rapid Capacity Strengthening of Rapid Response Teams in National and County Emergency Operations Center on the Outbreak Response Module						
Update the FETP frontline Epidemiology Course to include material on All Outbreak Response Module and National Disease Surveillance Repository.	**MOH IPs**	X	X			
Develop an inventory of Active and Reserve RRTs stationed Nationally and in Counties.		X	X	X		
Conduct a Training Needs Assessment among Active RRTs stationed within the Emergency Operations Center.			X			
Conduct Trainings to cover gaps in uptake of the Frontline FETP course and the new surveillance and outbreak response tools including All Outbreak Response Module and National Disease Surveillance Repository.			X	X		
Thematic Area 8: Digitise Public Health Emergency Operations Centers for effective detection of public health events, preparedness, and responses						
Focus Area 8.1: Adopt Incident Management software to strengthen response capabilities of national and regional Public Health Emergency Operations Centers (PHEOCs)	**MOH IPs**					
Establish an incident command center to streamline Public Health Emergency Operations.				X	X	X
Integrate data management systems to collect, analyse, and share data relevant to incident management.				X	X	X
Focus area 8.2: Adopt digital communication tools to strengthen collaboration and communication in PHEOCs.						
Establish a robust communication system.				X	X	X
Focus area 8.3: Enable establishment and activation of strategic stockpiles' rapid response to public health emergencies.						
Identify and implement a commodity management software.				X	X	X

5

Implementation & Scaleup

5.1. Logical Framework

The E-Public Health Architecture aims to achieve E-Surveillance system that supports disease surveillance activities broadly, accurately and in near real-time facilitating and enabling the One Health approach for surveillance of public health threats by all actors. The architecture will guide investments in the development of the National Disease Surveillance Repository, the outbreak response platform while strengthening routine surveillance and integration with the national information exchange. Strategic investments will be made in technology, health workforce, infrastructure, digital systems, and stakeholder management to ensure common approaches and synergy with existing MoH investments in digital health.



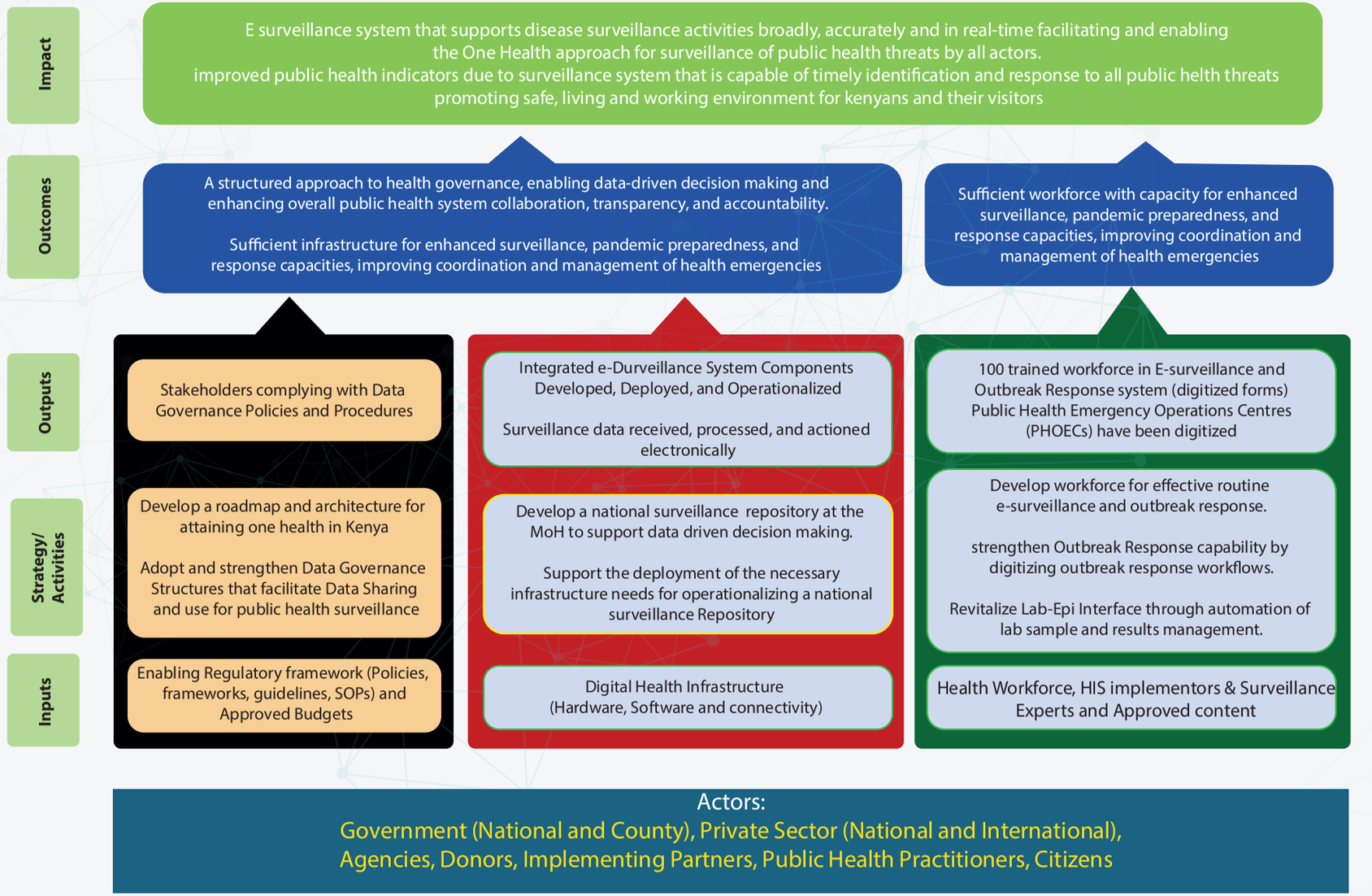


Figure 14: KNPHI Digital Public Health Implementation Logical Framework

6

Sustainability Plan



The execution of a sustainable digitization plan entails putting in place strategies to ensure the initiatives are economically, socially, and environmentally sustainable. While the strategy outlines key areas as part of its sustainability plan, all stakeholders will have a role to play in developing subject specific plans to align with what has been described herein. The key areas will be governance, and implementation of the envisioned e-surveillance architecture.

Category	Subcategory	Details
Governance	Health Data Governance	All surveillance data will be handled in line with the principles of data protection outlined in Kenya's Health Data Governance Framework. All stakeholders will be required to accurately describe how the principles of data protection have been adhered to.
	Alignment with Kenya's Digital Superhighway	The National Government and the Ministry of Health require all digital health systems to be part of the digital superhighway. All human health systems must undergo a certification framework to ensure adherence to standards and guidelines. The architecture allows for integration of data from other sectors for a robust surveillance ecosystem towards attaining the one health goal.
Infrastructure (Hardware, Software, and Connectivity)	Server Hosting	KNPHI will advocate for all applications/servers collecting and hosting surveillance data to be housed at the designated data center. This ensures that the institute remains the custodian of all respective applications and data. All partners/stakeholders involved will be guided by the Ministry through the hosting process and any requirements needed.
	Lifecycle Planning	KNPHI in collaboration with stakeholders, will develop plans outlining the entire lifecycle of hardware, including regular servicing, decommissioning, and replacement. This ensures adequate plans are in place to avoid downtimes due to wear and tear of key equipment.
	Cost Management	Software development and management should be effective, especially in resource-constrained environments. KNPHI will work with stakeholders to develop a clear budget and financial plan for digitization projects, including initial investments, ongoing maintenance, potential upgrades, decommissioning of obsolete technology, and data transfer/consolidation where applicable. Re-use or enhancement will be encouraged instead of building from scratch.
	Innovation, Adaptation, and Scalability	Digital systems should be designed with the most up-to-date technology, be flexible enough to scale up or down, and efficient to ensure improved performance in case of varying usage. All stakeholders are expected to develop solutions that can accommodate confined usage or mass usage where applicable. KNPHI will coordinate stakeholders and be part of the design and development process to ensure accommodation for all possible scenarios.

	Open Source and Shared Resources	The ministry encourages the utilization of open-source technologies and shared digital resources to reduce costs and enhance collaboration.
	Technical Documentation	Sustainable software development and management require proper documentation. The ministry advocates for all applications to be hosted in an MoH-approved code repository. Additionally, proper versioning of the applications is required with up-to-date documentation of the technology used in the development of the application.
Implementation	Implementation Planning and Monitoring	KNPHI will jointly work with all stakeholders to develop implementation plans that are SMART. Periodically, KNPHI with the help of the stakeholders will monitor the progress of implementation against the set targets and address any blockers where applicable.
	Skill Development	To meet workforce and stakeholder capacity needs, systems and data training will be implemented for the various levels of implementation in both electronic and physical formats. Additionally, all training materials will be hosted in the MOH learning academy for ease of accessibility.
	Partnerships and Collaborations	Where applicable, partnerships with various key organizations and agencies will be encouraged. The surveillance technical working group will provide oversight in key engagements and collaborations.

Table 4: Sustainability Plan

7

Risk Management Plan

Realisation of e-IDSR would be dependent on successful navigation of people, process, and technology risks. The following is a summary of key risks:

Risk Areas	Risk Level	Preventive/mitigation measures	Responsible
Systems & features not ready on time	High	Review system readiness and target to have a stop date on development	All Surveillance Stakeholders
Competing tasks	High	Prioritise on high impact systems and readiness	All Surveillance Stakeholders
EOCs lack a preferred tool	High	Plan engagement with EOC focal persons	All Surveillance Stakeholders
Resistance to change from healthcare providers	High	Conduct change management sessions.	All Surveillance Stakeholders
Data security and privacy concerns	High	Implement Security in technology, operations and governance layers. Use UPI	All Surveillance Stakeholders
Budget constraints	High	Secure funding through public-private partnerships	All Surveillance Stakeholders
Duplication of systems	High	Collapse system features	All Surveillance Stakeholders
Residual paper collection tools	High	Provide ability to have their data be hosted electronically	All Surveillance Stakeholders
Lack of terminology services	High	Adopt the NHDD terminology services	All Surveillance Stakeholders

Table 5: Risk Management Matrix

8

Appendices

8.1. Appendix 1: Systems and Tools in Use/ Supported by Various Surveillance Initiatives.

Disease Surveillance Initiative	Information System	Description
Port Health	• Traveler health surveillance form	• A digital platform that supports capture of case investigation and contact tracing information from travellers arriving in the country through designated ports of entry. • Digital format as Jitenge platform
	• Covid19 Vaccination register	• The registry containing records of persons vaccinated against of COVID-19. • It is used generates Covid19 vaccine certificates and serves as a central platform for authenticating vaccination status of Kenyan Citizens traveling outside the country. • Digital format as the M-Chanjo platform.
	• Travel Vaccinations Register	• A manual tool used to keep a record of individuals vaccinated in Kenya at the designated points of entry. • Commonly used for recording yellow fever, Polio vaccines administered. It is also used for Meningitis and Hepatitis vaccines.
	• Regional Electronic Cargo & Driver Tracking System	• A digital platform that was linked with selected COVID-19 testing laboratories. • It was established to support authentication Covid-19 test results and vaccination status of cargo drivers at the points of entry during the COVID-19 epidemic. • The system is currently not in active use.
	• PanaBIOS	• This system supported the issuance of trusted travel (TT) codes which were used as a means of ascertaining authenticity of travellers' Covid 19 certificates during the COVID-19 epidemic. • PanaBIOS was linked to trusted laboratories in the region. It is currently not in active use.

National Laboratory Services	• Labware LMIS	Manages lab processes
	• Lab test summary report form (MOH 706A & B)	Aggregation tools for all lab tests - Monthly
	• Laboratory testing and reporting form (MOH 227)	Sample Request form for tests
National Vaccines and Immunization Program	• Immunization Tally Sheet (MOH 702), • Immunization Summary Sheet (MOH 710) • Mother & child Health Booklet, • Immunization Monitor Chart, • Immunization Permanent Register • Defaulter tracking register	• Tools for recording, collating summarizing, and reporting routine immunization data across the country. • Paper based (Especially at facility)- Analysis done in excel, • Some EMR facilities have digital form of these tools.
	• Adverse Events Following Immunization reporting form	• A manual tool used for documenting surveillance of adverse events derived from vaccines.
	• Chanjo.ke (National, Regional, subcounty)	• A digital platform used for monitoring COVID-19 vaccine uptake.
Vector borne and Neglected Tropical Diseases	• WHO Data collection tools	

Table 6: Systems and Tools in Use/ Supported by Various Surveillance Initiatives

Zoonotic Disease Unit	• RVF ESS Tools • Active Outbreak Investigation Tool- Case Investigation Form • Vaccine use and Monitoring tools	
	• Sanitary report form	For field surveillance officers, private practitioners, meat inspectors, etc. to report animal diseases and events in real time.
	• Zero report form	Captures the absence of priority animal diseases every time a surveillance officer examines an animal.
	• Community diseases report form	Used by the community disease reporters to report observed clinical signs on real time basis.
	• Sample submission form	Used when practitioners (public and private) submit samples to the lab.
	• Livestock vaccinations form	Captures vaccination figures for all livestock diseases as they are undertaken at the ward level.
	Reporting Tools • KABS tool • DHIS2 • ARIS • WAHIS	• Kenya Animal Bio surveillance System • Aggregate Reporting and visualization platform • Animal Resource Information System • World Animal Health Information System
Environmental Surveillance	• Community Led Total Sanitation (CLTS) Real Time-Monitoring Information System	• Monitor households and community data.
	• Rural Urban Sanitation Hygiene (RUSH) RT-MIS	• Rural sanitation tracker.
	• MOH 708	• Public and Environmental health Services monthly summary form.

Table 6: Systems and Tools in Use/ Supported by Various Surveillance Initiatives

8.2. Appendix 2: Digital Tools Prioritised for First Phase of Transmission

Name of System/Tool	Data Collected	Status of Tool/System (Digitised/Manual)	System Owner
ChanjoKe	Covid- 19 Vaccination registry	Digital	MOH Kenya
KeNVIP repository	Immunization registry, Vaccines stocks and cold chain equipment, Smart learning, IEC materials, dashboard	Digital	JKUAT
KHIS 2	Aggregate program reporting	Digital	MOH Kenya
MOH 515 (Reporting monthly) - eCHIS	Events documentation at the community level	Digital	MOH Kenya
mDharura	Events documentation at the community level	Digital	ICAP
NYSS CBS **(NYSS is a Norwegian word. It means to get word or wind of** **something; to find out about something; to hear of a rumour.)**	Risk Assessment and reporting for true and false alerts	Digital	RED CROSS
EMRs/ TiBU LMIS	Captures clinical observations of patients	Digital	MOH/ Proprietary system owners
Web IFA	Environmental surveillance	Digital	WHO
EARS	Emergency Alerts reporting system for any notifiable diseases.	Digital	MOH Kenya
Kenya Animal Surveillance system (KAP)	Animal surveillance	Digital	MOH Kenya
Kenya Covid-19 Tracker	Validation of Covid- 19 vaccination	Digital	
Jitenge Plus	A digital platform that supports capture of case investigation and contact tracing information from travellers arriving in the country through designated ports of entry	Digital	MOH Kenya
LMIS	Manages lab processes	Digital	NPHL
Lab Manifest	Extract from MOH 502 lab section; accompanies samples to the lab	Digital	Health Facilities

Table 7: Digital Tools Prioritised for First Phase of Transmission

8.3. Appendix 3: Manual Tools Prioritised for Digitisation and Transmission

Name of System/Tool	MOH Number	Status of Tool/System (Digitised/Manual)	Owner of the System
Adverse Events Following Immunisation (AEFI) A&B - A-Event reporting form, B- Investigation form PvERS		Manual	MOH Kenya
IDSR line listing form	MOH 503	Manual	MOH Kenya
Monthly Summary Reporting Form	MOH 504		MOH Kenya
Weekly Summary Reporting Form	MOH 505	Manual	MOH Kenya
Travelers Surveillance Form	-	Manual	POE
Sample Request form	MOH 227	Manual	MOH

Table 8: Manual Tools Prioritised for Digitisation and Transmission

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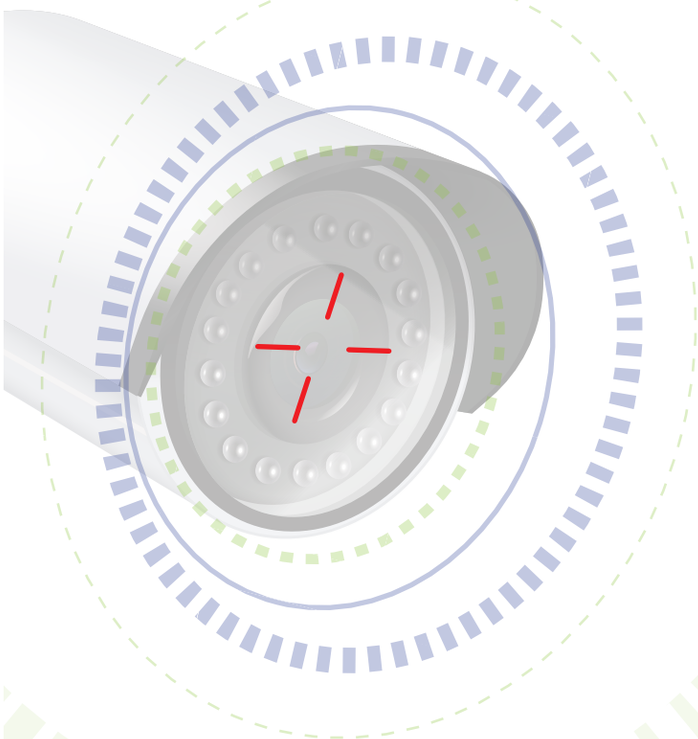
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